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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display all columns when printing DataFrames
pd.set_option('display.max_columns', None)

# Set a general plotting style
sns.set_style("whitegrid")

file_path =
"/content/drive/MyDrive/seasonal_agriculture_performance_dataset.csv"

df = pd.read_csv(file_path)

print("Dataset loaded successfully!")

print("\n===== FIRST 5 ROWS =====")
print(df.head())

print("\n===== LAST 5 ROWS =====")
print(df.tail())

print("\n===== NUMBER OF ROWS AND COLUMNS =====")
print(df.shape)

print("\n===== COLUMN NAMES =====")
print(df.columns.tolist())

print("\n===== DATA TYPES =====")
print(df.dtypes)

print("\n===== DATASET INFORMATION =====")
df.info()

print("\n===== DESCRIPTIVE STATISTICS =====")
print(df.describe())

print("\n===== DESCRIPTIVE STATISTICS INCLUDING CATEGORICAL COLUMNS =====")
print(df.describe(include='all'))

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print("\n===== MISSING VALUES =====")
print(df.isnull().sum())

print("\n===== MISSING VALUE PERCENTAGE =====")

missing_percentage = (
    df.isnull().sum() / len(df) * 100
).sort_values(ascending=False)

print(missing_percentage)

# Numerical columns with missing values
numeric_columns_to_fill = [
    'Rainfall_mm',
    'Soil_Moisture_pct',
    'Yield_Tonnes_Ha'
]

for column in numeric_columns_to_fill:
    df[column] = df[column].fillna(df[column].median())

print("\nMissing values handled successfully.")

print("\n===== MISSING VALUES AFTER CLEANING =====")
print(df.isnull().sum())

print("\n===== NUMBER OF DUPLICATES =====")
print(df.duplicated().sum())

# Remove duplicates if any
df = df.drop_duplicates()

print("\nDataset shape after removing duplicates:")
print(df.shape)

print("\n===== UNIQUE SEASONS =====")
print(df['Season'].unique())

print("\n===== UNIQUE CROPS =====")
print(df['Crop'].unique())

print("\n===== UNIQUE STATES =====")
print(df['State'].unique())

print("\n===== UNIQUE IRRIGATION METHODS =====")
print(df['Irrigation_Method'].unique())

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print("\n===== SEASON COUNT =====")
print(df['Season'].value_counts())

print("\n===== CROP COUNT =====")
print(df['Crop'].value_counts())

print("\n===== STATE COUNT =====")
print(df['State'].value_counts())

print("\n===== IRRIGATION METHOD COUNT =====")
print(df['Irrigation_Method'].value_counts())


plt.figure(figsize=(8, 5))

sns.countplot(
    data=df,
    x='Season'
)

plt.title('Number of Farms by Season')
plt.xlabel('Season')
plt.ylabel('Number of Farms')

plt.show()


season_yield = df.groupby('Season')['Yield_Tonnes_Ha'].mean()

print("\n===== AVERAGE YIELD BY SEASON =====")
print(season_yield.sort_values(ascending=False))


# Visualization
plt.figure(figsize=(8, 5))

sns.barplot(
    data=df,
    x='Season',
    y='Yield_Tonnes_Ha',
    estimator='mean'
)

plt.title('Average Yield by Season')
plt.xlabel('Season')

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plt.ylabel('Average Yield (Tonnes/Ha)')
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x='Season',
    y='Yield_Tonnes_Ha'
)

plt.title('Yield Distribution by Season')
plt.xlabel('Season')
plt.ylabel('Yield (Tonnes/Ha)')
plt.show()

season_production = df.groupby(
    'Season'
)['Production_Tonnes'].mean()

print("\n===== AVERAGE PRODUCTION BY SEASON =====")
print(season_production.sort_values(ascending=False))

# Visualization
plt.figure(figsize=(8, 5))
sns.barplot(
    data=df,
    x='Season',
    y='Production_Tonnes',
    estimator='mean'
)

plt.title('Average Production by Season')
plt.xlabel('Season')
plt.ylabel('Production (Tonnes)')
plt.show()

season_profit = df.groupby(
    'Season'
)['Profit_INR'].mean()

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```

print("\n===== AVERAGE PROFIT BY SEASON =====")
print(season_profit.sort_values(ascending=False))

plt.figure(figsize=(8, 5))

sns.barplot(
    data=df,
    x='Season',
    y='Profit_INR',
    estimator='mean'
)

plt.title('Average Profit by Season')
plt.xlabel('Season')
plt.ylabel('Average Profit (INR)')

plt.show()

season_financial = df.groupby('Season').agg({
    'Revenue_INR': 'mean',
    'Total_Cost_INR': 'mean',
    'Profit_INR': 'mean'
})

print("\n===== FINANCIAL PERFORMANCE BY SEASON =====")
print(season_financial)

season_financial.plot(
    kind='bar',
    figsize=(10, 6)
)

plt.title('Average Revenue, Cost and Profit by Season')
plt.xlabel('Season')
plt.ylabel('Amount (INR)')
plt.xticks(rotation=0)

plt.show()

crop_analysis = df.groupby('Crop').agg({

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        'Yield_Tonnes_Ha': 'mean',
        'Production_Tonnes': 'mean',
        'Profit_INR': 'mean'
    })

    print("\n===== CROP-WISE PERFORMANCE =====")
    print(crop_analysis.sort_values(
        'Profit_INR',
        ascending=False
    ))

plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x='Crop',
    y='Profit_INR',
    estimator='mean'
)

plt.title('Average Profit by Crop')
plt.xlabel('Crop')
plt.ylabel('Average Profit (INR)')

plt.xticks(rotation=45)

plt.show()

plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x='Crop',
    y='Yield_Tonnes_Ha',
    estimator='mean'
)

plt.title('Average Yield by Crop')
plt.xlabel('Crop')
plt.ylabel('Average Yield (Tonnes/Ha)')

plt.xticks(rotation=45)

plt.show()

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state_analysis = df.groupby('State').agg({
    'Yield_Tonnes_Ha': 'mean',
    'Production_Tonnes': 'mean',
    'Profit_INR': 'mean'
})

print("\n===== STATE-WISE PERFORMANCE =====")
print(state_analysis.sort_values(
    'Profit_INR',
    ascending=False
))

plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x='State',
    y='Profit_INR',
    estimator='mean'
)

plt.title('Average Profit by State')
plt.xlabel('State')
plt.ylabel('Average Profit (INR)')

plt.xticks(rotation=45)

plt.show()

water_usage = df.groupby(
    'Season'
)['Water_Used_m3'].mean()

print("\n===== AVERAGE WATER USAGE BY SEASON =====")
print(water_usage.sort_values(ascending=False))

plt.figure(figsize=(8, 5))

sns.barplot(
    data=df,
    x='Season',
    y='Water_Used_m3',
    estimator='mean'
)

```

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plt.title('Average Water Usage by Season')
plt.xlabel('Season')
plt.ylabel('Water Used (m³)')

plt.show()

water_efficiency = df.groupby(
    'Crop'
)['Water_Efficiency_t_per_1000m3'].mean()

print("\n===== WATER EFFICIENCY BY CROP =====")
print(water_efficiency.sort_values(ascending=False))

plt.figure(figsize=(12, 6))

sns.barplot(
    x=water_efficiency.index,
    y=water_efficiency.values
)

plt.title('Average Water Efficiency by Crop')
plt.xlabel('Crop')
plt.ylabel('Tonnes per 1000 m³')

plt.xticks(rotation=45)

plt.show()

disease_risk = df.groupby(
    'Season'
)['Disease_Pest_Risk_pct'].mean()

print("\n===== AVERAGE DISEASE/PEST RISK BY SEASON =====")
print(disease_risk.sort_values(ascending=False))

plt.figure(figsize=(8, 5))

sns.barplot(
    data=df,
    x='Season',
    y='Disease_Pest_Risk_pct',
    estimator='mean'
)

plt.title('Average Disease/Pest Risk by Season')
plt.xlabel('Season')

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plt.ylabel('Disease/Pest Risk (%)')
plt.show()

rainfall = df.groupby(
    'Season'
)['Rainfall_mm'].mean()

print("\n===== AVERAGE RAINFALL BY SEASON =====")
print(rainfall.sort_values(ascending=False))

plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x='Season',
    y='Rainfall_mm'
)

plt.title('Rainfall Distribution by Season')
plt.xlabel('Season')
plt.ylabel('Rainfall (mm)')
plt.show()

temperature = df.groupby(
    'Season'
)['Avg_Temperature_C'].mean()

print("\n===== AVERAGE TEMPERATURE BY SEASON =====")
print(temperature.sort_values(ascending=False))

plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x='Season',
    y='Avg_Temperature_C'
)

plt.title('Temperature Distribution by Season')
plt.xlabel('Season')

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plt.ylabel('Temperature (°C)')
plt.show()

irrigation_count = df[
    'Irrigation_Method'
].value_counts()

print("\n===== IRRIGATION METHOD COUNT =====")
print(irrigation_count)

plt.figure(figsize=(9, 5))
sns.countplot(
    data=df,
    x='Irrigation_Method'
)

plt.title('Distribution of Irrigation Methods')
plt.xlabel('Irrigation Method')
plt.ylabel('Number of Farms')
plt.xticks(rotation=30)
plt.show()

irrigation_yield = df.groupby(
    'Irrigation_Method'
)['Yield_Tonnes_Ha'].mean()

print("\n===== AVERAGE YIELD BY IRRIGATION METHOD =====")
print(irrigation_yield.sort_values(ascending=False))

plt.figure(figsize=(10, 5))
sns.barplot(
    data=df,
    x='Irrigation_Method',
    y='Yield_Tonnes_Ha',
    estimator='mean'
)

plt.title('Average Yield by Irrigation Method')

```

```
plt.xlabel('Irrigation Method')
plt.ylabel('Average Yield (Tonnes/Ha)')

plt.xticks(rotation=30)

plt.show()
```

```
plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=df,
    x='Fertilizer_kg_ha',
    y='Yield_Tonnes_Ha',
    hue='Season'
)

plt.title('Fertilizer Usage vs Yield')
plt.xlabel('Fertilizer Used (kg/Ha)')
plt.ylabel('Yield (Tonnes/Ha)')

plt.show()
```

```
plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=df,
    x='Rainfall_mm',
    y='Yield_Tonnes_Ha',
    hue='Season'
)

plt.title('Rainfall vs Yield')
plt.xlabel('Rainfall (mm)')
plt.ylabel('Yield (Tonnes/Ha)')

plt.show()
```

```
plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=df,
    x='Avg_Temperature_C',
    y='Yield_Tonnes_Ha',
    hue='Season'
)
```

```
plt.title('Temperature vs Yield')
plt.xlabel('Average Temperature (°C)')
plt.ylabel('Yield (Tonnes/Ha)')
```

```
plt.show()
```

```
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(
    data=df,
    x='Yield_Tonnes_Ha',
    y='Profit_INR',
    hue='Season'
)
```

```
plt.title('Yield vs Profit by Season')
plt.xlabel('Yield (Tonnes/Ha)')
plt.ylabel('Profit (INR)')
```

```
plt.show()
```

```
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(
    data=df,
    x='Production_Tonnes',
    y='Profit_INR',
    hue='Season'
)
```

```
plt.title('Production vs Profit')
plt.xlabel('Production (Tonnes)')
plt.ylabel('Profit (INR)')
```

```
plt.show()
```

```
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(
    data=df,
    x='Revenue_INR',
    y='Profit_INR',
    hue='Season'
)
```

```

plt.title('Revenue vs Profit')
plt.xlabel('Revenue (INR)')
plt.ylabel('Profit (INR)')

plt.show()

numeric_df = df.select_dtypes(
    include=np.number
)

correlation = numeric_df.corr()

print("\n===== CORRELATION MATRIX =====")
print(correlation)

plt.figure(figsize=(16, 12))

sns.heatmap(
    correlation,
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    linewidths=0.5
)

plt.title('Correlation Heatmap')

plt.show()

profit_correlation = correlation[
    'Profit_INR'
].sort_values(ascending=False)

print("\n===== CORRELATION WITH PROFIT =====")
print(profit_correlation)

top_farms = df.sort_values(
    'Profit_INR',
    ascending=False
).head(10)

print("\n===== TOP 10 FARMS BY PROFIT =====")
print(

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top_farms[
    [
        'Farm_ID',
        'State',
        'Crop',
        'Season',
        'Profit_INR'
    ]
]

bottom_farms = df.sort_values(
    'Profit_INR'
).head(10)

print("\n===== BOTTOM 10 FARMS BY PROFIT =====")

print(
    bottom_farms[
        [
            'Farm_ID',
            'State',
            'Crop',
            'Season',
            'Profit_INR'
        ]
    ]
)

plt.figure(figsize=(10, 5))

sns.boxplot(
    data=df,
    x='Profit_INR'
)

plt.title('Profit Distribution and Outliers')
plt.xlabel('Profit (INR)')

plt.show()

plt.figure(figsize=(10, 5))

sns.boxplot(
    data=df,
    x='Yield_Tonnes_Ha'
)

```

```

)

plt.title('Yield Distribution and Outliers')
plt.xlabel('Yield (Tonnes/Ha)')

plt.show()

season_summary = df.groupby('Season').agg(
    Farms=('Farm_ID', 'count'),
    Avg_Yield=('Yield_Tonnes_Ha', 'mean'),
    Avg_Production=('Production_Tonnes', 'mean'),
    Avg_Revenue=('Revenue_INR', 'mean'),
    Avg_Cost=('Total_Cost_INR', 'mean'),
    Avg_Profit=('Profit_INR', 'mean'),
    Avg_Water_Used=('Water_Used_m3', 'mean'),
    Avg_Water_Efficiency=(
        'Water_Efficiency_t_per_1000m3',
        'mean'
    ),
    Avg_Disease_Risk=(
        'Disease_Pest_Risk_pct',
        'mean'
    )
).round(2)

print("\n===== COMPLETE SEASONAL SUMMARY =====")
print(season_summary)

best_season = season_summary[
    'Avg_Profit'
].idxmax()

best_season_profit = season_summary[
    'Avg_Profit'
].max()

print("\n===== BEST PERFORMING SEASON =====")

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print("Best Season:", best_season)
print("Average Profit:", best_season_profit)

worst_season = season_summary[
    'Avg_Profit'
].idxmin()

worst_season_profit = season_summary[
    'Avg_Profit'
].min()

print("\n===== LOWEST PERFORMING SEASON =====")
print("Lowest Season:", worst_season)
print("Average Profit:", worst_season_profit)

crop_profit = df.groupby(
    'Crop'
)['Profit_INR'].mean()

best_crop = crop_profit.idxmax()
best_crop_profit = crop_profit.max()

print("\n===== MOST PROFITABLE CROP =====")
print("Crop:", best_crop)
print("Average Profit:", best_crop_profit)

crop_yield = df.groupby(
    'Crop'
)['Yield_Tonnes_Ha'].mean()

best_yield_crop = crop_yield.idxmax()
best_yield = crop_yield.max()

print("\n===== HIGHEST YIELD CROP =====")
print("Crop:", best_yield_crop)
print("Average Yield:", best_yield)

state_profit = df.groupby(
    'State'
)['Profit_INR'].mean()

best_state = state_profit.idxmax()

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best_state_profit = state_profit.max()

print("\n===== HIGHEST PROFIT STATE =====")
print("State:", best_state)
print("Average Profit:", best_state_profit)

crop_water_efficiency = df.groupby(
    'Crop'
)['Water_Efficiency_t_per_1000m3'].mean()

best_water_crop = crop_water_efficiency.idxmax()
best_water_efficiency = crop_water_efficiency.max()

print("\n===== MOST WATER-EFFICIENT CROP =====")
print("Crop:", best_water_crop)
print("Water Efficiency:", best_water_efficiency)

season_risk = df.groupby(
    'Season'
)['Disease_Pest_Risk_pct'].mean()

highest_risk_season = season_risk.idxmax()
highest_risk = season_risk.max()

print("\n===== HIGHEST DISEASE/PEST RISK SEASON =====")
print("Season:", highest_risk_season)
print("Average Risk:", highest_risk)

season_water = df.groupby(
    'Season'
)['Water_Used_m3'].mean()

highest_water_season = season_water.idxmax()
highest_water_usage = season_water.max()

print("\n===== HIGHEST WATER USAGE SEASON =====")
print("Season:", highest_water_season)
print("Average Water Usage:", highest_water_usage)

print("\n")
print("=" * 60)
print("FINAL KEY FINDINGS")

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print("=" * 60)

print(
    f"\n1. Best performing season based on profit: "
    f"{best_season}"
)

print(
    f"    Average profit: ₹{best_season_profit:,.2f}"
)

print(
    f"\n2. Lowest performing season based on profit: "
    f"{worst_season}"
)

print(
    f"    Average profit: ₹{worst_season_profit:,.2f}"
)

print(
    f"\n3. Most profitable crop: {best_crop}"
)

print(
    f"    Average profit: ₹{best_crop_profit:,.2f}"
)

print(
    f"\n4. Highest yield crop: {best_yield_crop}"
)

print(
    f"    Average yield: {best_yield:.2f} tonnes/ha"
)

print(
    f"\n5. Highest profit state: {best_state}"
)

print(
    f"    Average profit: ₹{best_state_profit:,.2f}"
)

print(
    f"\n6. Most water-efficient crop: {best_water_crop}"
)

print(
    f"    Water efficiency: {best_water_efficiency:.2f}"
)
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)

print(
    f"\n7. Highest disease/pest risk season: "
    f"{highest_risk_season}"
)

print(
    f"    Average risk: {highest_risk:.2f}%"
)

print(
    f"\n8. Highest water usage season: "
    f"{highest_water_season}"
)

print(
    f"    Average water usage: {highest_water_usage:,.2f} m³"
)

print("\n")
print("=" * 60)
print("RECOMMENDATIONS")
print("=" * 60)

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season_summary.to_csv(
    "seasonal_summary.csv"
)

```

```

crop_analysis.to_csv(
    "crop_analysis.csv"
)

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```

state_analysis.to_csv(
    "state_analysis.csv"
)

```

```

print("\nAnalysis completed successfully!")
print("Summary files have been saved as CSV files.")

```

Dataset loaded successfully!

===== FIRST 5 ROWS =====

Farm_ID	State	District	Crop	Season
Farm_Area_Hectares \				
0_SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif
0.53				

1	SF10002	Maharashtra	Nalgonda	Maize	Kharif
6.53					
2	SF10003	Telangana	Warangal	Pulses	Rabi
4.86					
3	SF10004	Telangana	Indore	Rice	Kharif
4.50					
4	SF10005	Karnataka	Ludhiana	Maize	Zaid
4.21					

	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day
Soil_pH \				
0	486.4	24.3	69.1	5.8
7.14				
1	855.4	25.7	78.4	5.1
5.20				
2	455.6	23.9	56.4	10.3
5.82				
3	753.2	29.7	65.2	6.3
5.98				
4	101.6	34.1	55.2	6.5
6.90				

	Soil_Moisture_pct	Nitrogen_kg_ha	Phosphorus_kg_ha
Potassium_kg_ha \			
0	32.8	75.8	61.3
91.0			
1	32.0	82.1	35.4
121.8			
2	8.1	118.5	51.5
117.9			
3	23.7	138.6	55.6
115.5			
4	29.3	124.1	69.0
121.4			

	Irrigation_Method	Fertilizer_kg_ha	Pesticide_Litre_ha
Seed_Quality_Score \			
0	Drip	242.7	4.33
0.76			
1	Flood	271.4	6.86
0.94			
2	Drip	162.6	6.08
0.98			
3	Rainfed	182.2	6.39
0.74			
4	Flood	244.9	7.61
0.90			

	Yield_Tonnes_Ha	Production_Tonnes	Market_Price_INR_Tonne
Total_Cost_INR \			

0	2.46	1.30	23700
42662			
1	0.30	1.96	20613
492351			
2	0.52	2.53	75283
315210			
3	1.84	8.28	24244
296444			
4	2.21	9.30	20818
337959			

	Revenue_INR	Profit_INR	Water_Used_m3
Water_Efficiency_t_per_1000m3 \			
0	30810	-11852	237
5.485			
1	40401	-451950	4953
0.396			
2	190466	-124744	1275
1.984			
3	200740	-95704	3834
2.160			
4	193607	-144352	3287
2.829			

	Disease_Pest_Risk_pct
0	55.3
1	49.9
2	33.7
3	50.6
4	32.3

===== LAST 5 ROWS =====

	Farm_ID	State	District	Crop	Season
Farm_Area_Hectares \					
3995	SF13996	Karnataka	Rajkot	Wheat	Kharif
4.72					
3996	SF13997	Karnataka	Krishna	Chilli	Kharif
8.77					
3997	SF13998	Andhra Pradesh	Erode	Pulses	Zaid
10.47					
3998	SF13999	Punjab	Warangal	Cotton	Kharif
1.73					
3999	SF14000	Punjab	Nashik	Wheat	Kharif
3.31					

	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day
\				
3995	865.4	27.3	51.4	6.3
3996	947.5	29.2	85.5	6.1

3997	80.0	35.4	68.5	8.6
3998	475.0	28.2	61.7	6.4
3999	984.3	25.8	77.2	5.9
	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_ha	Phosphorus_kg_ha \
3995	7.97	27.3	104.3	44.8
3996	6.57	33.7	127.8	44.9
3997	7.25	21.1	126.7	59.9
3998	6.11	30.5	107.5	20.7
3999	6.16	35.7	137.1	92.6
	Potassium_kg_ha	Irrigation_Method	Fertilizer_kg_ha	
	Pesticide_Litre_ha \			
3995	94.0	Drip	271.1	
5.45				
3996	91.3	Drip	176.6	
4.99				
3997	148.0	Drip	234.7	
6.27				
3998	91.0	Sprinkler	329.7	
2.60				
3999	145.5	Sprinkler	205.9	
6.72				
	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	\
3995	0.74	0.51	2.41	
3996	0.80	2.75	24.12	
3997	0.99	0.62	6.49	
3998	0.98	1.35	2.34	
3999	0.99	2.36	7.81	
	Market_Price_INR_Tonne	Total_Cost_INR	Revenue_INR	Profit_INR
\				
3995	22717	362917	54748	-308169
3996	106459	496471	2567791	2071320
3997	66860	785632	433921	-351711
3998	75282	153904	176160	22256
3999	28812	263872	225022	-38850
	Water_Used_m3	Water_Efficiency_t_per_1000m3		
Disease_Pest_Risk_pct				
3995	2016		1.195	

45.2		
3996	5351	4.508
59.7		
3997	3392	1.913
29.2		
3998	1220	1.918
50.4		
3999	1800	4.339
62.3		

===== NUMBER OF ROWS AND COLUMNS =====
(4000, 28)

===== COLUMN NAMES =====

```
[ 'Farm_ID', 'State', 'District', 'Crop', 'Season',
'Farm_Area_Hectares', 'Rainfall_mm', 'Avg_Temperature_C',
'Humidity_pct', 'Sunlight_Hours_Day', 'Soil_pH', 'Soil_Moisture_pct',
'Nitrogen_kg_ha', 'Phosphorus_kg_ha', 'Potassium_kg_ha',
'Irrigation_Method', 'Fertilizer_kg_ha', 'Pesticide_Litre_ha',
'Seed_Quality_Score', 'Yield_Tonnes_Ha', 'Production_Tonnes',
'Market_Price_INR_Tonne', 'Total_Cost_INR', 'Revenue_INR',
'Profit_INR', 'Water_Used_m3', 'Water_Efficiency_t_per_1000m3',
'Disease_Pest_Risk_pct']
```

===== DATA TYPES =====

Farm_ID	object
State	object
District	object
Crop	object
Season	object
Farm_Area_Hectares	float64
Rainfall_mm	float64
Avg_Temperature_C	float64
Humidity_pct	float64
Sunlight_Hours_Day	float64
Soil_pH	float64
Soil_Moisture_pct	float64
Nitrogen_kg_ha	float64
Phosphorus_kg_ha	float64
Potassium_kg_ha	float64
Irrigation_Method	object
Fertilizer_kg_ha	float64
Pesticide_Litre_ha	float64
Seed_Quality_Score	float64
Yield_Tonnes_Ha	float64
Production_Tonnes	float64
Market_Price_INR_Tonne	int64
Total_Cost_INR	int64
Revenue_INR	int64
Profit_INR	int64

```
Water_Used_m3                int64
Water_Efficiency_t_per_1000m3 float64
Disease_Pest_Risk_pct        float64
dtype: object
```

===== DATASET INFORMATION =====

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4000 entries, 0 to 3999
```

```
Data columns (total 28 columns):
```

#	Column	Non-Null Count	Dtype
0	Farm_ID	4000 non-null	object
1	State	4000 non-null	object
2	District	4000 non-null	object
3	Crop	4000 non-null	object
4	Season	4000 non-null	object
5	Farm_Area_Hectares	4000 non-null	float64
6	Rainfall_mm	3952 non-null	float64
7	Avg_Temperature_C	4000 non-null	float64
8	Humidity_pct	4000 non-null	float64
9	Sunlight_Hours_Day	4000 non-null	float64
10	Soil_pH	4000 non-null	float64
11	Soil_Moisture_pct	3960 non-null	float64
12	Nitrogen_kg_ha	4000 non-null	float64
13	Phosphorus_kg_ha	4000 non-null	float64
14	Potassium_kg_ha	4000 non-null	float64
15	Irrigation_Method	4000 non-null	object
16	Fertilizer_kg_ha	4000 non-null	float64
17	Pesticide_Litre_ha	4000 non-null	float64
18	Seed_Quality_Score	4000 non-null	float64
19	Yield_Tonnes_Ha	3968 non-null	float64
20	Production_Tonnes	4000 non-null	float64
21	Market_Price_INR_Tonne	4000 non-null	int64
22	Total_Cost_INR	4000 non-null	int64
23	Revenue_INR	4000 non-null	int64
24	Profit_INR	4000 non-null	int64
25	Water_Used_m3	4000 non-null	int64
26	Water_Efficiency_t_per_1000m3	4000 non-null	float64
27	Disease_Pest_Risk_pct	4000 non-null	float64

```
dtypes: float64(17), int64(5), object(6)
```

```
memory usage: 875.1+ KB
```

===== DESCRIPTIVE STATISTICS =====

	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C
Humidity_pct \			
count	4000.000000	3952.000000	4000.000000
mean	7.888265	601.152657	26.81890
63.210500			

std	4.223764	288.925025	3.81818
11.960834			
min	0.500000	80.000000	16.20000
25.000000			
25%	4.260000	370.200000	23.90000
54.775000			
50%	7.895000	582.000000	26.90000
63.000000			
75%	11.650000	834.950000	29.60000
71.900000			
max	15.000000	1395.200000	39.70000
95.000000			

	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct
Nitrogen_kg_ha \			
count	4000.000000	4000.000000	3960.000000
4000.000000			
mean	7.323050	6.701010	26.508232
120.250825			
std	1.121186	0.641461	6.712484
31.216452			
min	3.500000	5.200000	8.000000
40.000000			
25%	6.600000	6.260000	21.700000
98.500000			
50%	7.300000	6.690000	26.400000
120.100000			
75%	8.100000	7.140000	31.300000
141.225000			
max	11.000000	8.400000	47.000000
220.000000			

	Phosphorus_kg_ha	Potassium_kg_ha	Fertilizer_kg_ha \
count	4000.000000	4000.000000	4000.000000
mean	57.781850	104.760050	186.036000
std	17.015074	27.742681	60.260583
min	15.000000	30.000000	40.000000
25%	46.200000	86.200000	143.975000
50%	57.500000	105.100000	186.250000
75%	69.400000	123.600000	227.200000
max	120.000000	200.000000	400.000000

	Pesticide_Litre_ha	Seed_Quality_Score	Yield_Tonnes_Ha \
count	4000.000000	4000.000000	3968.000000
mean	5.070935	0.826138	5.270595
std	1.978879	0.101747	13.361398
min	0.500000	0.650000	0.300000
25%	3.690000	0.740000	1.050000
50%	5.120000	0.830000	1.740000
75%	6.400000	0.910000	2.840000

max	12.080000	1.000000	101.440000		
	Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	\	
count	4000.000000	4000.000000	4.000000e+03		
mean	43.246210	44714.166250	5.263036e+05		
std	126.880521	30266.161922	2.944191e+05		
min	0.150000	2594.000000	2.682600e+04		
25%	5.020000	21429.250000	2.745775e+05		
50%	11.770000	25913.000000	5.190825e+05		
75%	24.355000	68279.000000	7.619965e+05		
max	1424.400000	131240.000000	1.349990e+06		
	Revenue_INR	Profit_INR	Water_Used_m3	\	
count	4.000000e+03	4.000000e+03	4000.000000		
mean	6.378600e+05	1.115565e+05	6045.570250		
std	6.369747e+05	5.450819e+05	5572.944306		
min	6.107000e+03	-1.019223e+06	128.000000		
25%	2.075172e+05	-1.487318e+05	2268.750000		
50%	4.568305e+05	3.673000e+03	4435.000000		
75%	8.368748e+05	2.161878e+05	7829.750000		
max	5.080900e+06	4.345421e+06	40902.000000		
	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct			
count	4000.000000	4000.000000			
mean	5.385214	46.365875			
std	8.838101	12.424417			
min	0.135000	5.000000			
25%	1.661000	37.300000			
50%	2.937500	46.400000			
75%	5.150250	54.900000			
max	79.800000	88.900000			
===== DESCRIPTIVE STATISTICS INCLUDING CATEGORICAL COLUMNS					
=====					
	Farm_ID	State	District	Crop	Season
Farm_Area_Hectares	\				
count	4000	4000	4000	4000	4000
4000.000000					
unique	4000	8	10	8	3
NaN					
top	SF13984	Andhra Pradesh	Ludhiana	Rice	Kharif
NaN					
freq	1	529	427	690	1779
NaN					
mean	NaN	NaN	NaN	NaN	NaN
7.888265					
std	NaN	NaN	NaN	NaN	NaN
4.223764					
min	NaN	NaN	NaN	NaN	NaN
0.500000					

25%	NaN	NaN	NaN	NaN	NaN
4.260000					
50%	NaN	NaN	NaN	NaN	NaN
7.895000					
75%	NaN	NaN	NaN	NaN	NaN
11.650000					
max	NaN	NaN	NaN	NaN	NaN
15.000000					

	Rainfall_mm	Avg_Temperature_C	Humidity_pct
Sunlight_Hours_Day \			
count	3952.000000	4000.000000	4000.000000
4000.000000			
unique	NaN	NaN	NaN
NaN			
top	NaN	NaN	NaN
NaN			
freq	NaN	NaN	NaN
NaN			
mean	601.152657	26.81890	63.210500
7.323050			
std	288.925025	3.81818	11.960834
1.121186			
min	80.000000	16.20000	25.000000
3.500000			
25%	370.200000	23.90000	54.775000
6.600000			
50%	582.000000	26.90000	63.000000
7.300000			
75%	834.950000	29.60000	71.900000
8.100000			
max	1395.200000	39.70000	95.000000
11.000000			

	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_ha
Phosphorus_kg_ha \			
count	4000.000000	3960.000000	4000.000000
4000.000000			
unique	NaN	NaN	NaN
NaN			
top	NaN	NaN	NaN
NaN			
freq	NaN	NaN	NaN
NaN			
mean	6.701010	26.508232	120.250825
57.781850			
std	0.641461	6.712484	31.216452
17.015074			
min	5.200000	8.000000	40.000000

15.000000			
25%	6.260000	21.700000	98.500000
46.200000			
50%	6.690000	26.400000	120.100000
57.500000			
75%	7.140000	31.300000	141.225000
69.400000			
max	8.400000	47.000000	220.000000
120.000000			

	Potassium_kg_ha	Irrigation_Method	Fertilizer_kg_ha	\
count	4000.000000	4000	4000.000000	
unique	NaN	4	NaN	
top	NaN	Flood	NaN	
freq	NaN	1310	NaN	
mean	104.760050	NaN	186.036000	
std	27.742681	NaN	60.260583	
min	30.000000	NaN	40.000000	
25%	86.200000	NaN	143.975000	
50%	105.100000	NaN	186.250000	
75%	123.600000	NaN	227.200000	
max	200.000000	NaN	400.000000	

	Pesticide_Litre_ha	Seed_Quality_Score	Yield_Tonnes_Ha	\
count	4000.000000	4000.000000	3968.000000	
unique	NaN	NaN	NaN	
top	NaN	NaN	NaN	
freq	NaN	NaN	NaN	
mean	5.070935	0.826138	5.270595	
std	1.978879	0.101747	13.361398	
min	0.500000	0.650000	0.300000	
25%	3.690000	0.740000	1.050000	
50%	5.120000	0.830000	1.740000	
75%	6.400000	0.910000	2.840000	
max	12.080000	1.000000	101.440000	

	Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	\
count	4000.000000	4000.000000	4.000000e+03	
unique	NaN	NaN	NaN	
top	NaN	NaN	NaN	
freq	NaN	NaN	NaN	
mean	43.246210	44714.166250	5.263036e+05	
std	126.880521	30266.161922	2.944191e+05	
min	0.150000	2594.000000	2.682600e+04	
25%	5.020000	21429.250000	2.745775e+05	
50%	11.770000	25913.000000	5.190825e+05	
75%	24.355000	68279.000000	7.619965e+05	
max	1424.400000	131240.000000	1.349990e+06	

Revenue_INR	Profit_INR	Water_Used_m3	\
-------------	------------	---------------	---

count	4.000000e+03	4.000000e+03	4000.000000
unique	NaN	NaN	NaN
top	NaN	NaN	NaN
freq	NaN	NaN	NaN
mean	6.378600e+05	1.115565e+05	6045.570250
std	6.369747e+05	5.450819e+05	5572.944306
min	6.107000e+03	-1.019223e+06	128.000000
25%	2.075172e+05	-1.487318e+05	2268.750000
50%	4.568305e+05	3.673000e+03	4435.000000
75%	8.368748e+05	2.161878e+05	7829.750000
max	5.080900e+06	4.345421e+06	40902.000000

	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct
count	4000.000000	4000.000000
unique	NaN	NaN
top	NaN	NaN
freq	NaN	NaN
mean	5.385214	46.365875
std	8.838101	12.424417
min	0.135000	5.000000
25%	1.661000	37.300000
50%	2.937500	46.400000
75%	5.150250	54.900000
max	79.800000	88.900000

===== MISSING VALUES =====

Farm_ID	0
State	0
District	0
Crop	0
Season	0
Farm_Area_Hectares	0
Rainfall_mm	48
Avg_Temperature_C	0
Humidity_pct	0
Sunlight_Hours_Day	0
Soil_pH	0
Soil_Moisture_pct	40
Nitrogen_kg_ha	0
Phosphorus_kg_ha	0
Potassium_kg_ha	0
Irrigation_Method	0
Fertilizer_kg_ha	0
Pesticide_Litre_ha	0
Seed_Quality_Score	0
Yield_Tonnes_Ha	32
Production_Tonnes	0
Market_Price_INR_Tonne	0
Total_Cost_INR	0
Revenue_INR	0

```

Profit_INR                                0
Water_Used_m3                             0
Water_Efficiency_t_per_1000m3             0
Disease_Pest_Risk_pct                     0
dtype: int64

===== MISSING VALUE PERCENTAGE =====
Rainfall_mm                               1.2
Soil_Moisture_pct                          1.0
Yield_Tonnes_Ha                           0.8
Farm_ID                                    0.0
Crop                                        0.0
District                                   0.0
Farm_Area_Hectares                         0.0
Season                                     0.0
Avg_Temperature_C                         0.0
Humidity_pct                              0.0
Sunlight_Hours_Day                        0.0
State                                      0.0
Soil_pH                                    0.0
Nitrogen_kg_ha                            0.0
Potassium_kg_ha                           0.0
Phosphorus_kg_ha                         0.0
Irrigation_Method                        0.0
Fertilizer_kg_ha                         0.0
Pesticide_Litre_ha                       0.0
Seed_Quality_Score                       0.0
Production_Tonnes                        0.0
Market_Price_INR_Tonne                   0.0
Total_Cost_INR                           0.0
Revenue_INR                              0.0
Profit_INR                               0.0
Water_Used_m3                             0.0
Water_Efficiency_t_per_1000m3             0.0
Disease_Pest_Risk_pct                     0.0
dtype: float64

```

Missing values handled successfully.

```

===== MISSING VALUES AFTER CLEANING =====
Farm_ID                                    0
State                                      0
District                                   0
Crop                                        0
Season                                     0
Farm_Area_Hectares                         0
Rainfall_mm                               0
Avg_Temperature_C                         0
Humidity_pct                              0
Sunlight_Hours_Day                        0

```

```

Soil_pH                0
Soil_Moisture_pct      0
Nitrogen_kg_ha         0
Phosphorus_kg_ha       0
Potassium_kg_ha        0
Irrigation_Method      0
Fertilizer_kg_ha       0
Pesticide_Litre_ha     0
Seed_Quality_Score     0
Yield_Tonnes_Ha        0
Production_Tonnes      0
Market_Price_INR_Tonne 0
Total_Cost_INR         0
Revenue_INR            0
Profit_INR             0
Water_Used_m3          0
Water_Efficiency_t_per_1000m3 0
Disease_Pest_Risk_pct  0
dtype: int64

```

```
===== NUMBER OF DUPLICATES =====
```

```
0
```

```

Dataset shape after removing duplicates:
(4000, 28)

```

```
===== UNIQUE SEASONS =====
```

```
['Kharif' 'Rabi' 'Zaid']
```

```
===== UNIQUE CROPS =====
```

```
['Wheat' 'Maize' 'Pulses' 'Rice' 'Cotton' 'Chilli' 'Groundnut'
'Sugarcane']
```

```
===== UNIQUE STATES =====
```

```
['Andhra Pradesh' 'Maharashtra' 'Telangana' 'Karnataka' 'Gujarat'
'Tamil Nadu' 'Punjab' 'Madhya Pradesh']
```

```
===== UNIQUE IRRIGATION METHODS =====
```

```
['Drip' 'Flood' 'Rainfed' 'Sprinkler']
```

```
===== SEASON COUNT =====
```

```

Season
Kharif    1779
Rabi      1627
Zaid       594
Name: count, dtype: int64

```

```
===== CROP COUNT =====
```

```

Crop
Rice      690

```

Wheat	614
Maize	551
Cotton	508
Pulses	496
Groundnut	424
Chilli	412
Sugarcane	305

Name: count, dtype: int64

===== STATE COUNT =====

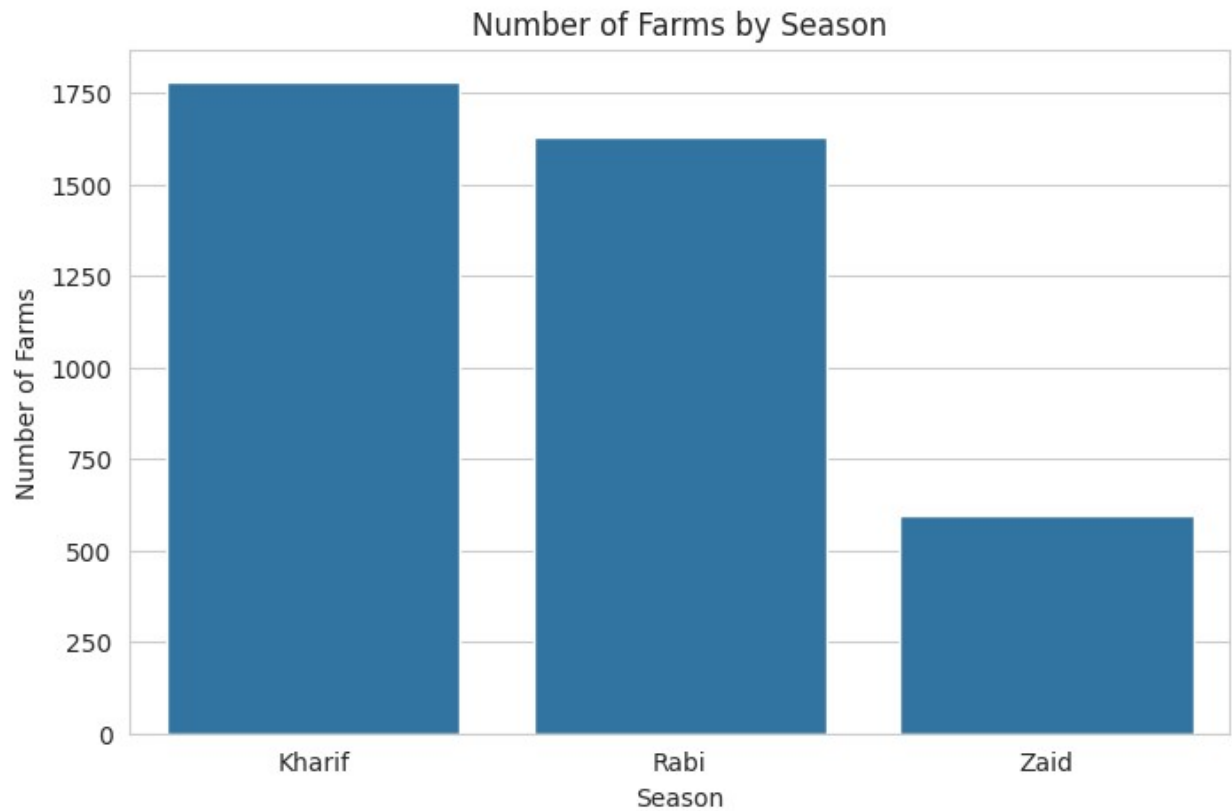
State	
Andhra Pradesh	529
Telangana	516
Maharashtra	512
Madhya Pradesh	497
Karnataka	489
Gujarat	488
Tamil Nadu	485
Punjab	484

Name: count, dtype: int64

===== IRRIGATION METHOD COUNT =====

Irrigation_Method	
Flood	1310
Rainfed	1041
Drip	915
Sprinkler	734

Name: count, dtype: int64



===== AVERAGE YIELD BY SEASON =====

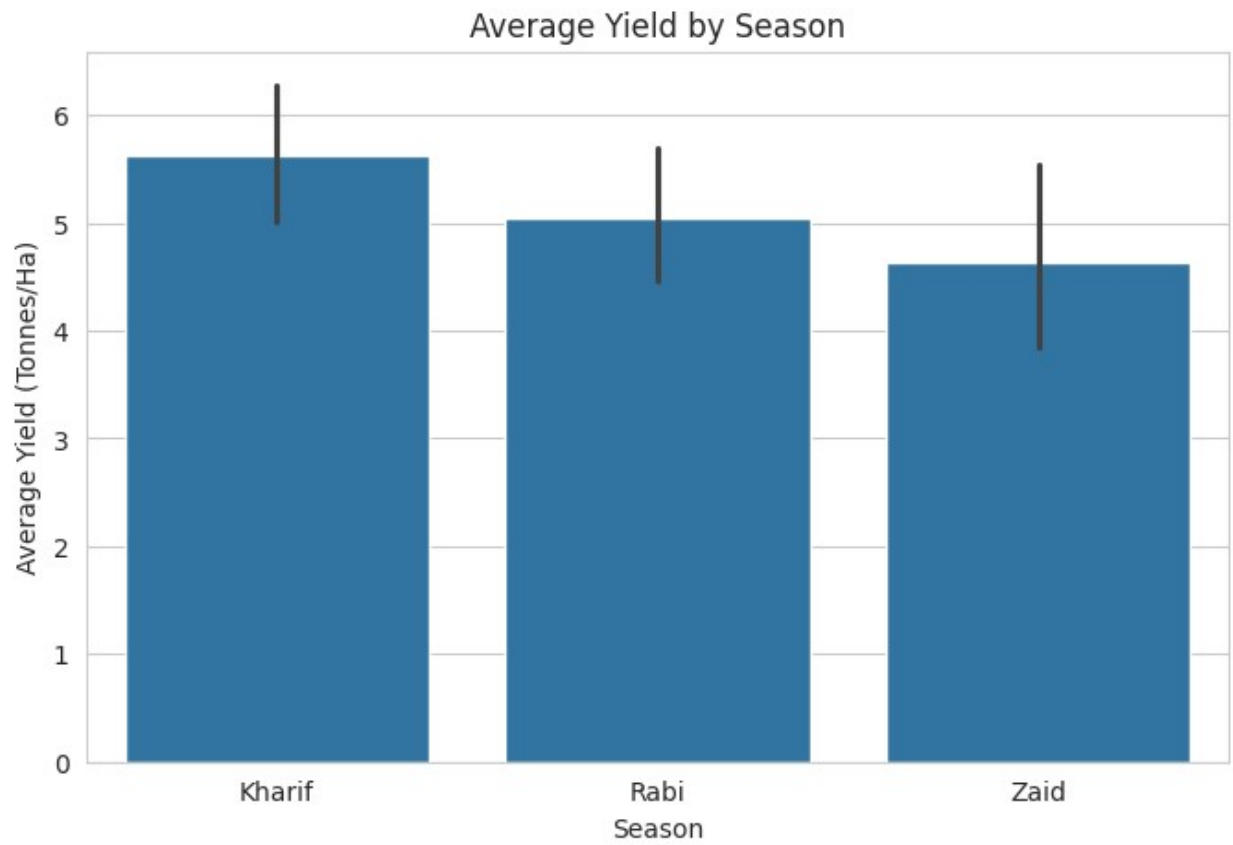
Season

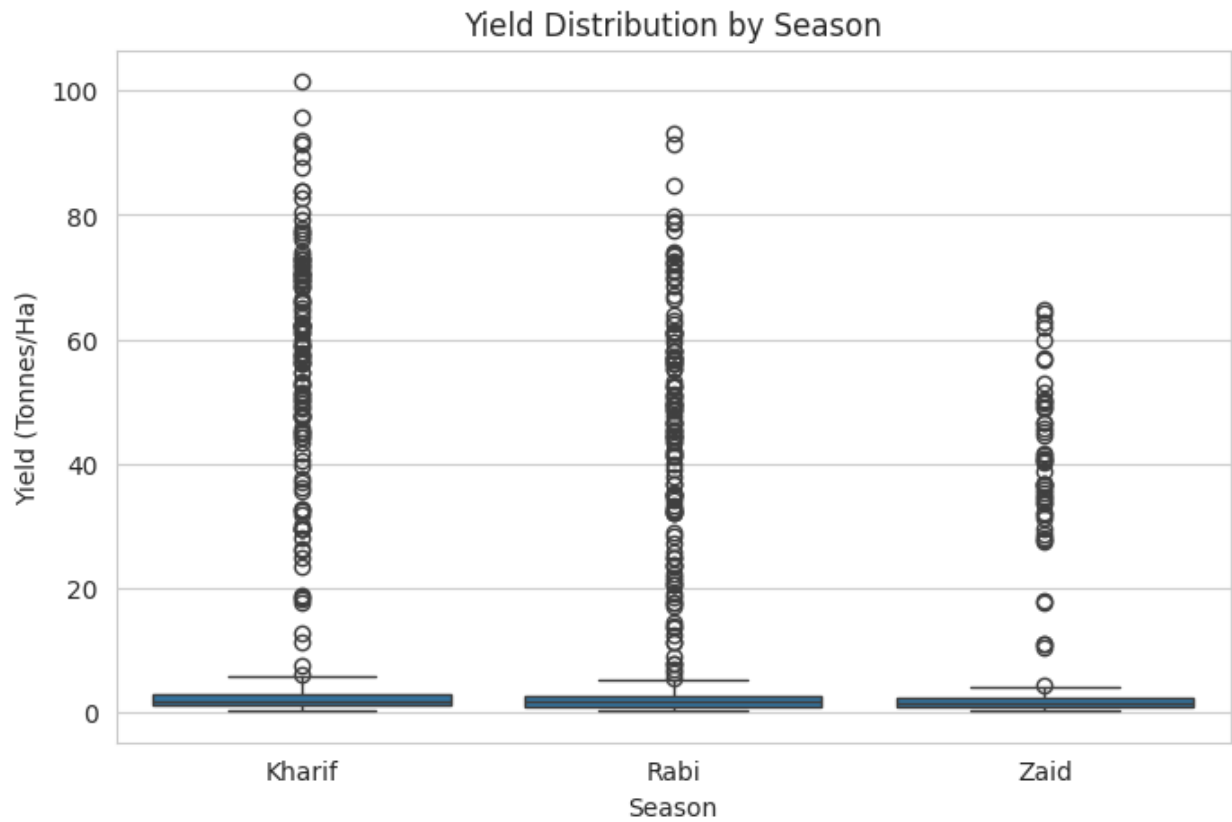
Kharif 5.628612

Rabi 5.039435

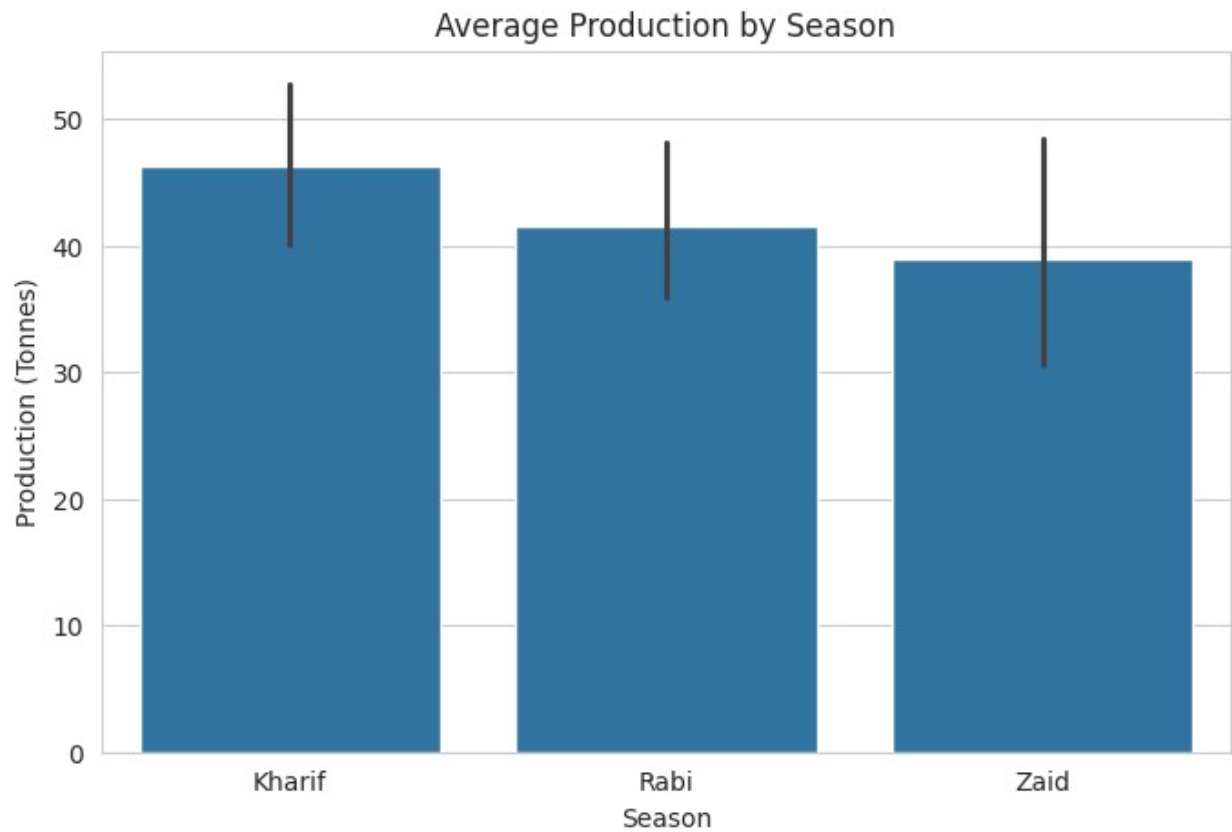
Zaid 4.641313

Name: Yield_Tonnes_Ha, dtype: float64





```
===== AVERAGE PRODUCTION BY SEASON =====  
Season  
Kharif    46.311192  
Rabi      41.486712  
Zaid      38.886111  
Name: Production_Tonnes, dtype: float64
```



===== AVERAGE PROFIT BY SEASON =====

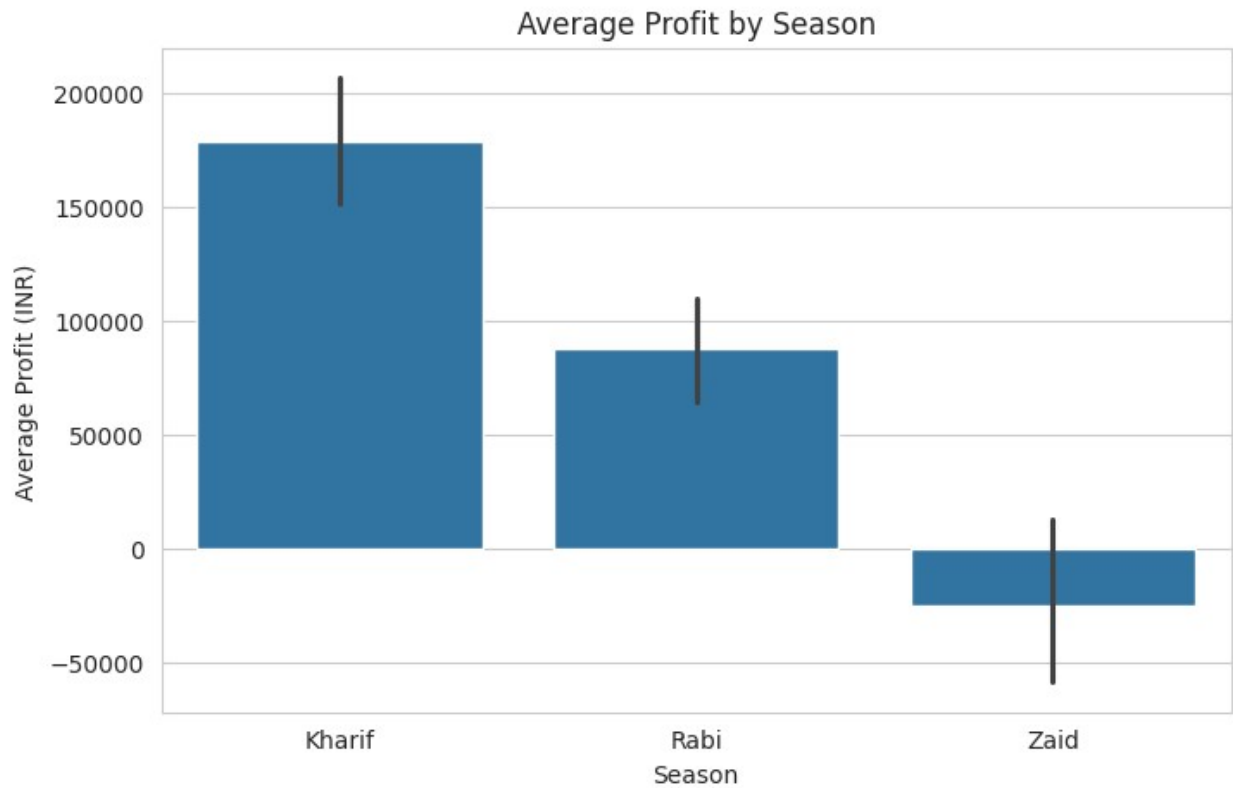
Season

Kharif 178914.647555

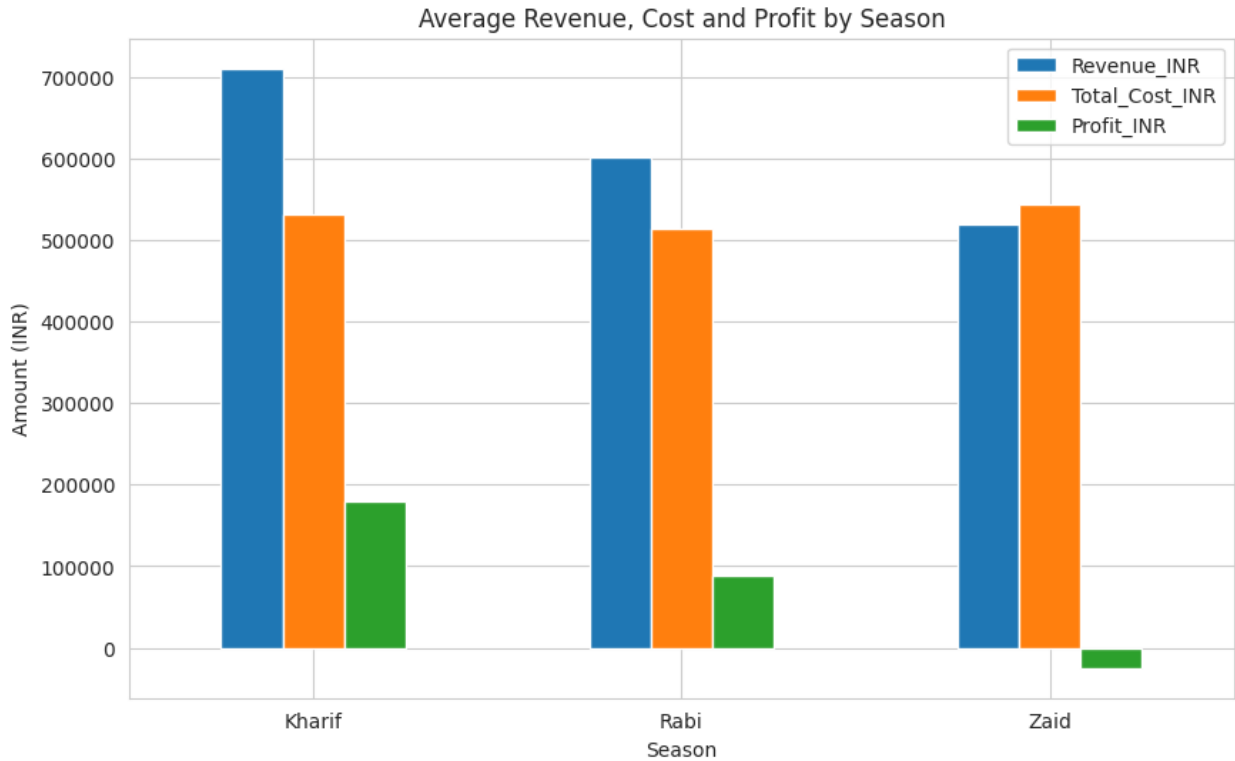
Rabi 87689.470191

Zaid -24804.824916

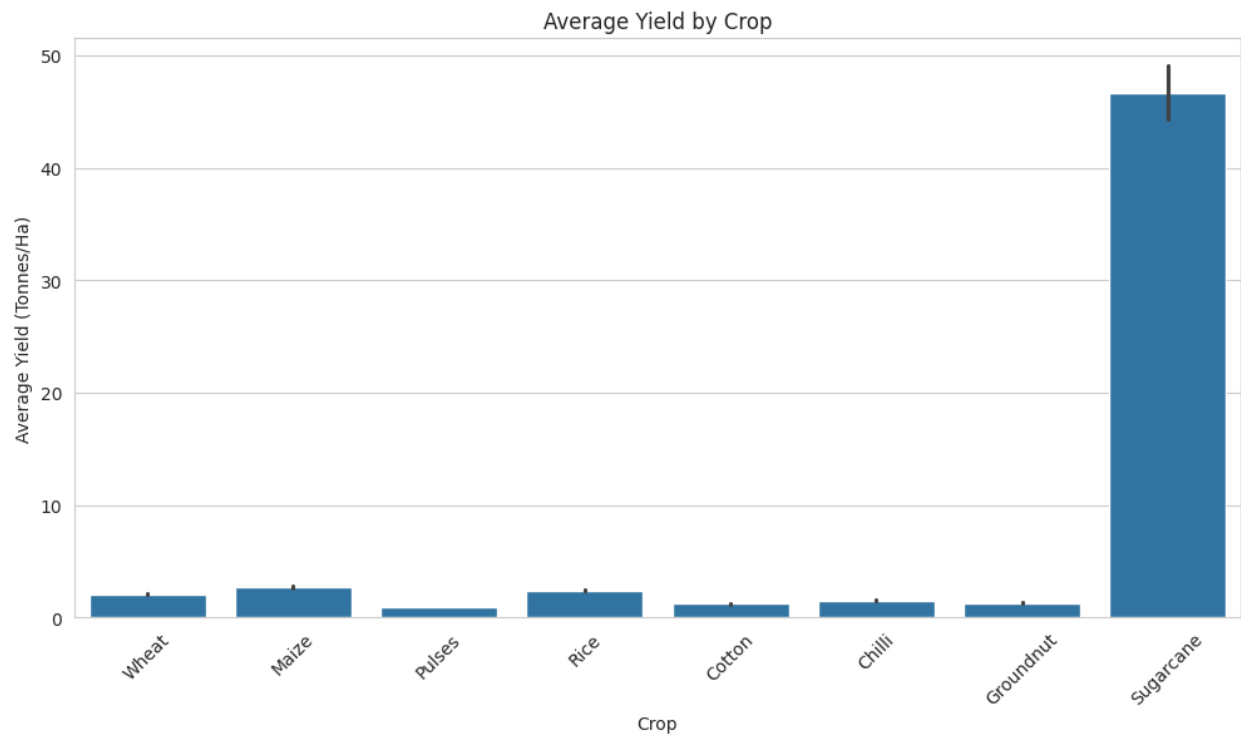
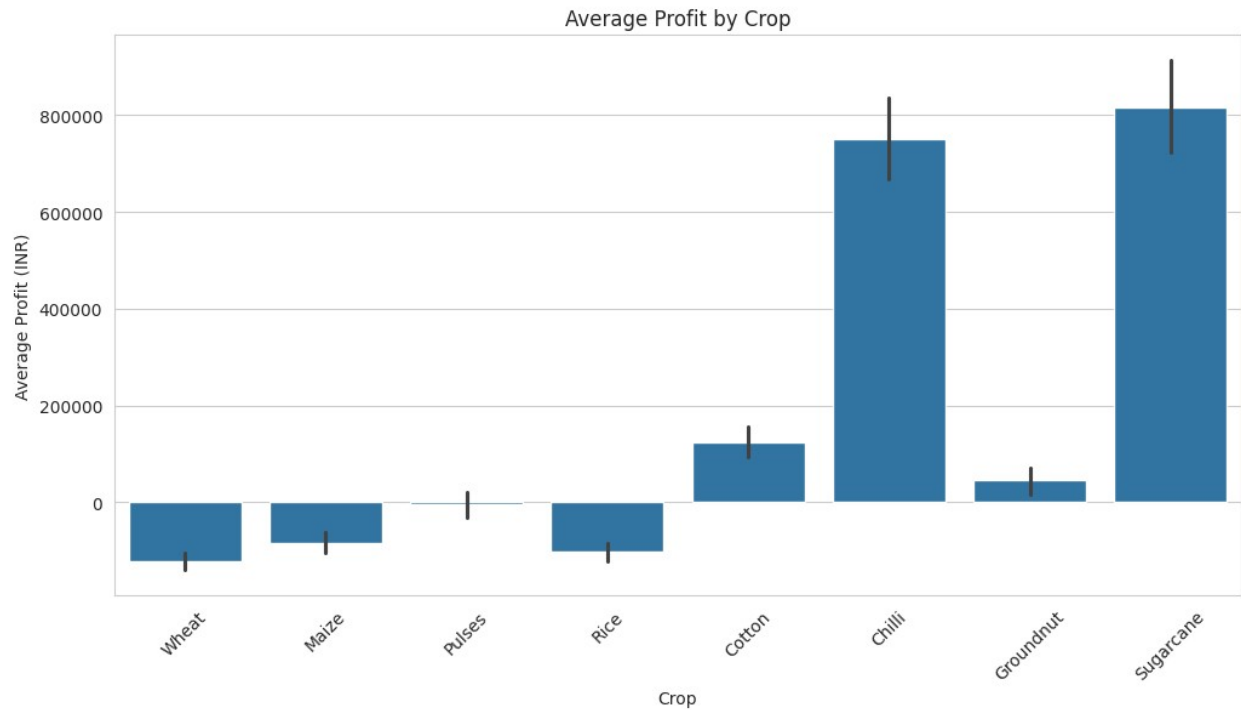
Name: Profit_INR, dtype: float64



===== FINANCIAL PERFORMANCE BY SEASON =====			
	Revenue_INR	Total_Cost_INR	Profit_INR
Season			
Kharif	710719.055649	531804.408094	178914.647555
Rabi	601526.048556	513836.578365	87689.470191
Zaid	519171.904040	543976.728956	-24804.824916



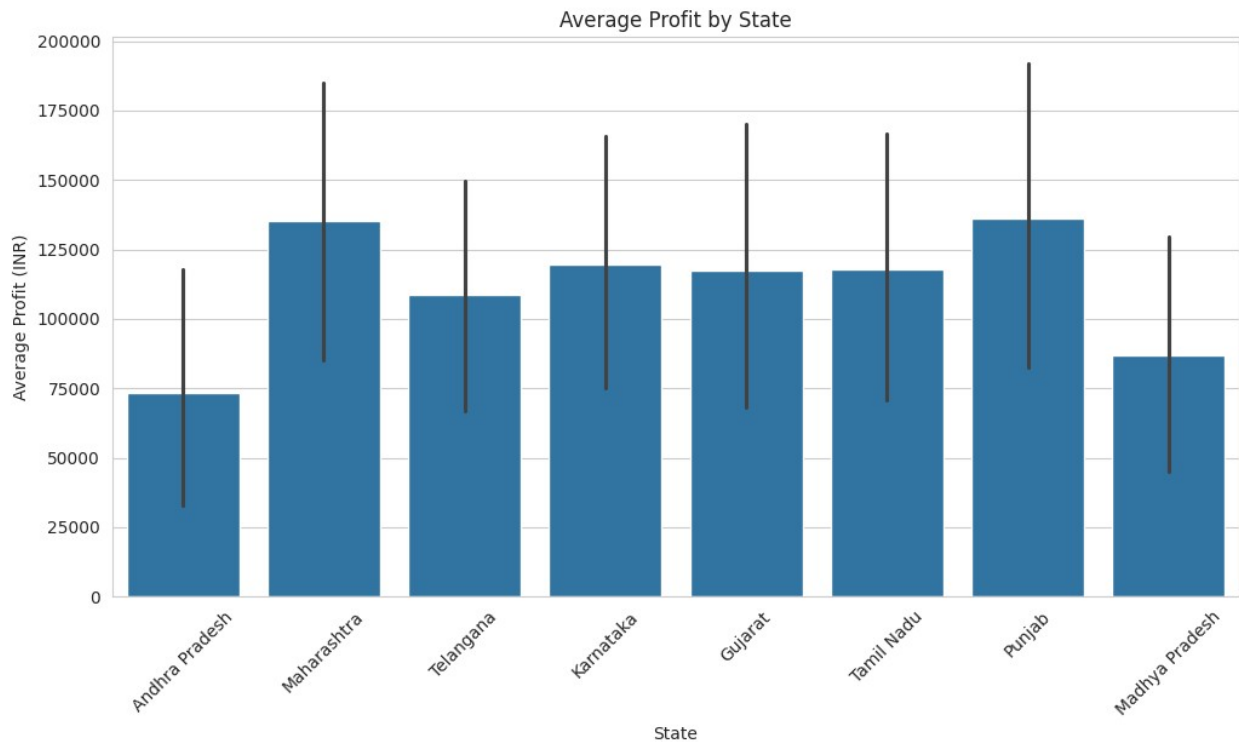
===== CROP-WISE PERFORMANCE =====			
Crop	Yield_Tonnes_Ha	Production_Tonnes	Profit_INR
Sugarcane	46.642361	392.413016	817187.986885
Chilli	1.542767	12.382888	750878.342233
Cotton	1.229783	9.395866	124546.915354
Groundnut	1.322901	9.655825	44858.120283
Pulses	0.921512	7.401411	-4238.052419
Maize	2.710036	21.781125	-83978.326679
Rice	2.431304	19.288623	-102213.504348
Wheat	2.108013	16.853909	-123398.337134



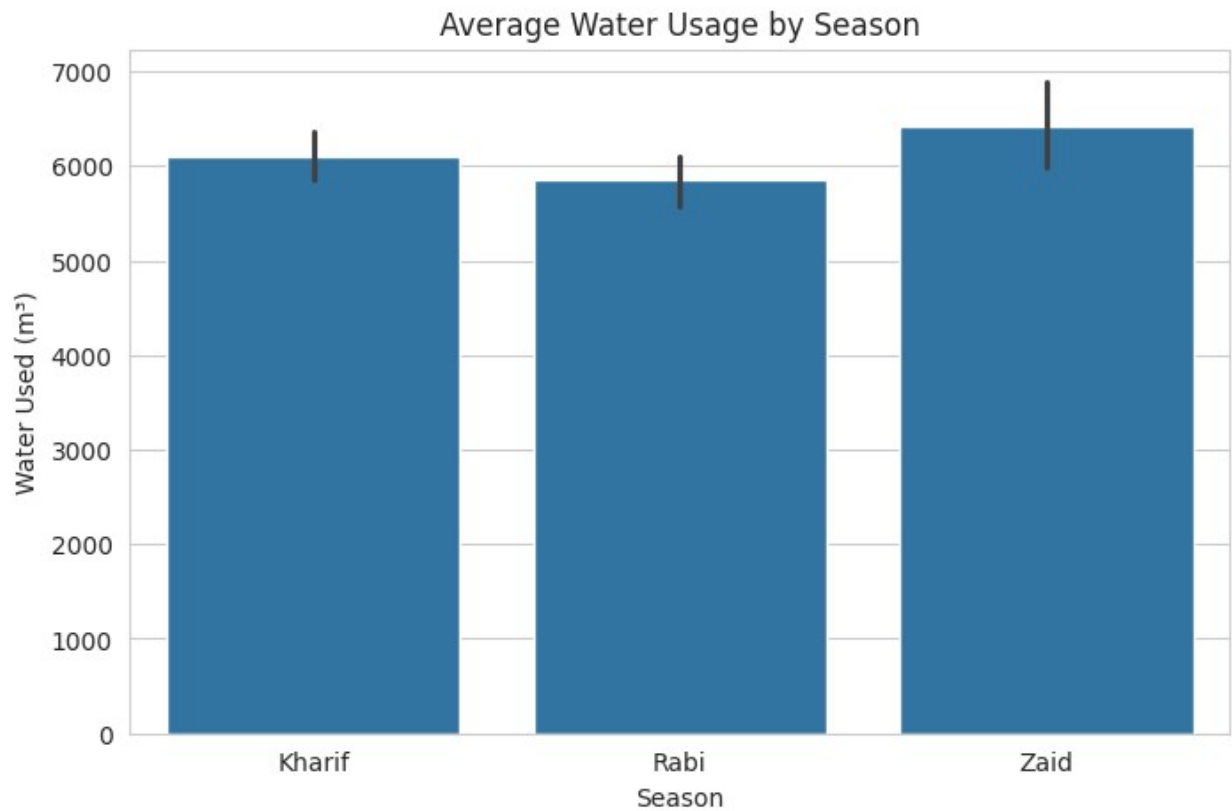
===== STATE-WISE PERFORMANCE =====

State	Yield_Tonnes_Ha	Production_Tonnes	Profit_INR
Punjab	6.121116	52.233099	136025.636364

Maharashtra	5.017480	43.984043	135428.855469
Karnataka	5.685481	47.732720	119422.335378
Tamil Nadu	4.875175	40.840371	117744.748454
Gujarat	5.659016	48.331988	117568.235656
Telangana	5.044477	39.254186	108829.616279
Madhya Pradesh	4.989940	37.559517	86840.756539
Andhra Pradesh	4.628771	36.913157	73453.527410



```
===== AVERAGE WATER USAGE BY SEASON =====
Season
Zaid      6419.893939
Kharif    6102.201237
Rabi      5846.987093
Name: Water_Used_m3, dtype: float64
```



===== WATER EFFICIENCY BY CROP =====

Crop

Sugarcane 30.477672

Maize 5.602995

Wheat 4.628393

Groundnut 3.154795

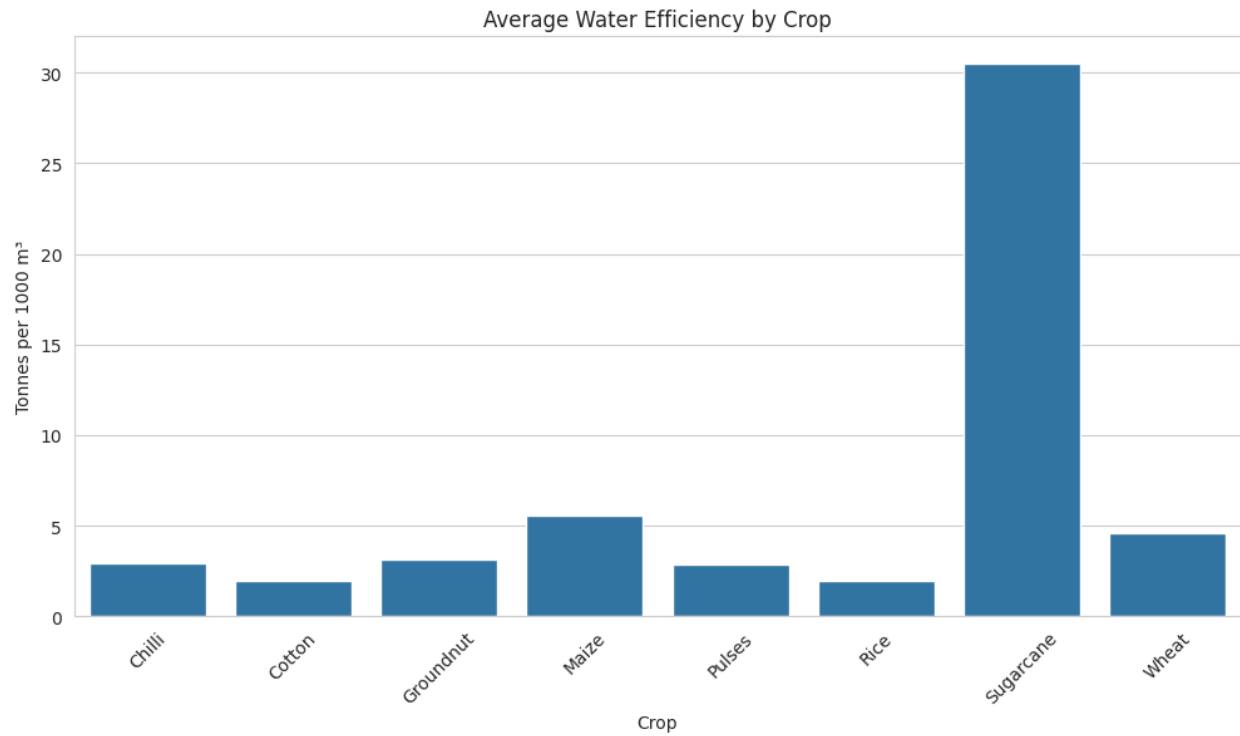
Chilli 2.947709

Pulses 2.896504

Rice 1.946112

Cotton 1.938016

Name: Water_Efficiency_t_per_1000m3, dtype: float64



===== AVERAGE DISEASE/PEST RISK BY SEASON =====

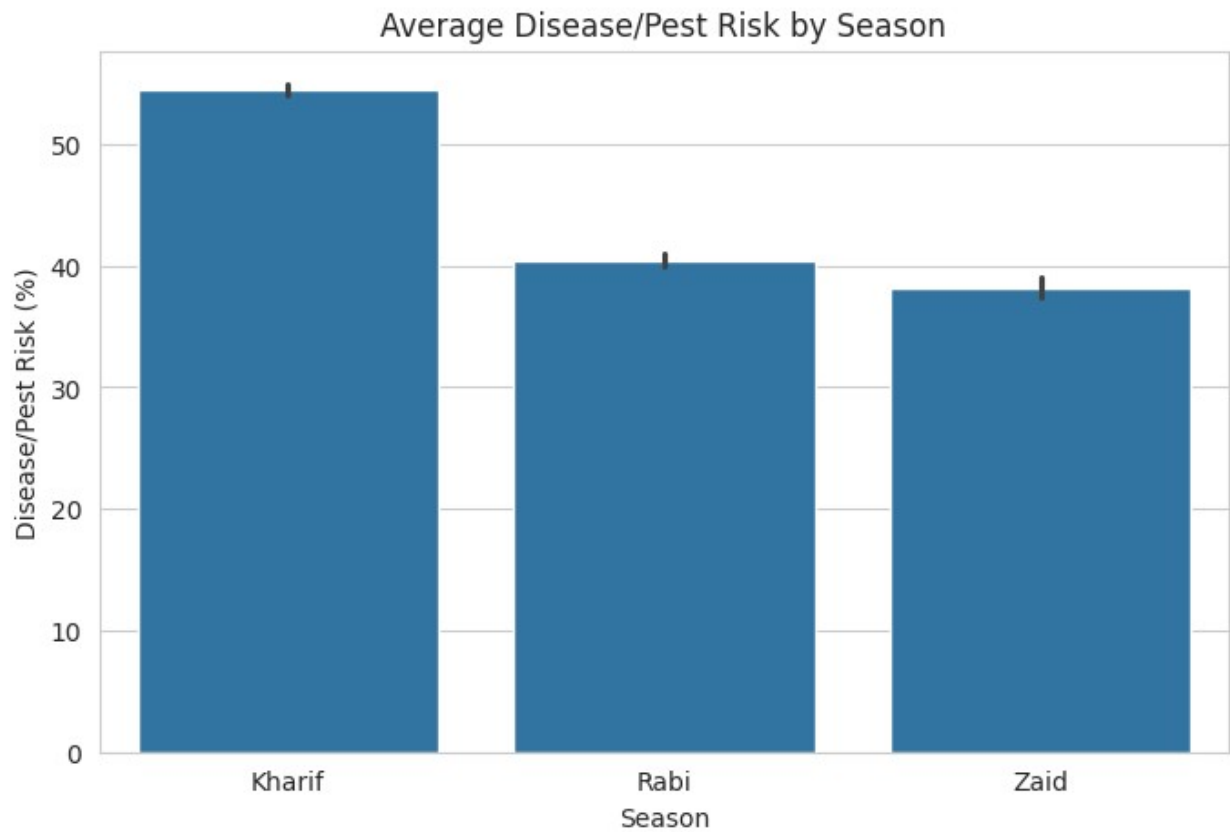
Season

Khariif 54.467903

Rabi 40.482114

Zaid 38.216667

Name: Disease_Pest_Risk_pct, dtype: float64



===== AVERAGE RAINFALL BY SEASON =====

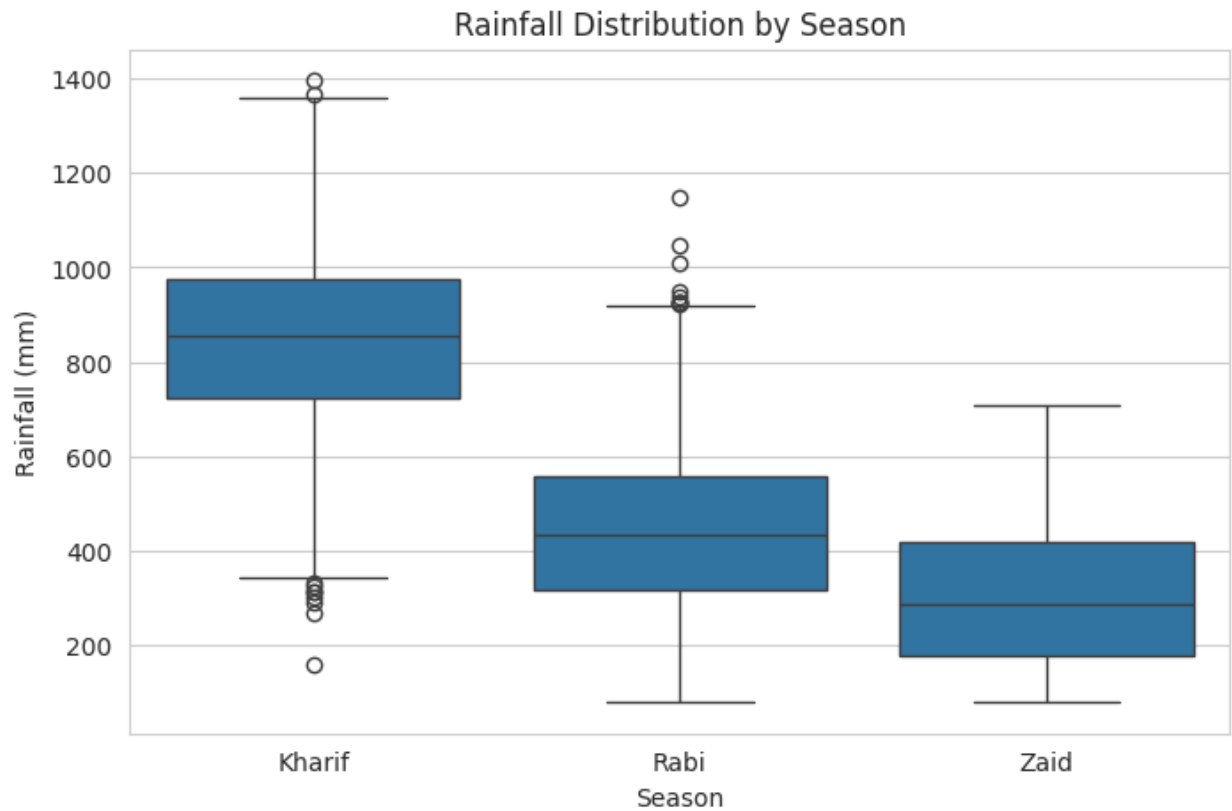
Season

Kharif 849.199663

Rabi 437.615366

Zaid 304.654714

Name: Rainfall_mm, dtype: float64



===== AVERAGE TEMPERATURE BY SEASON =====

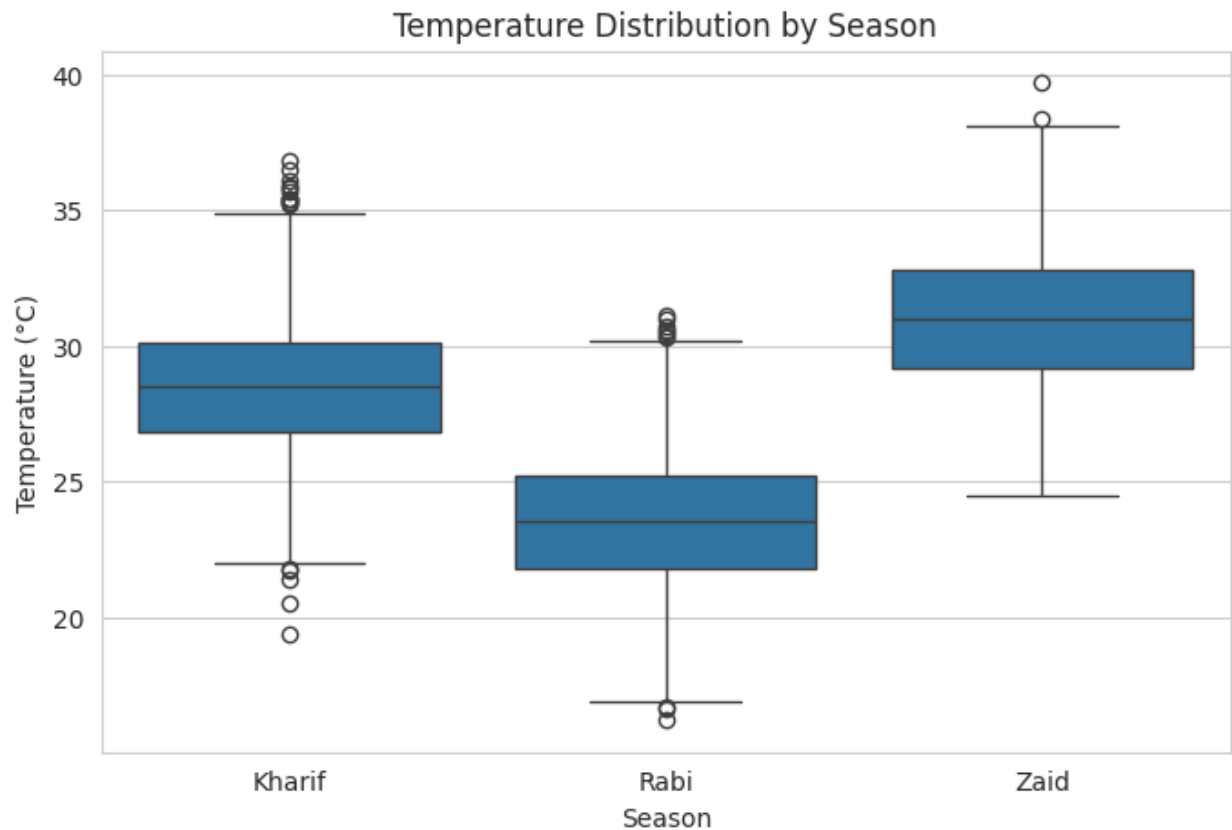
Season

Zaid 31.042088

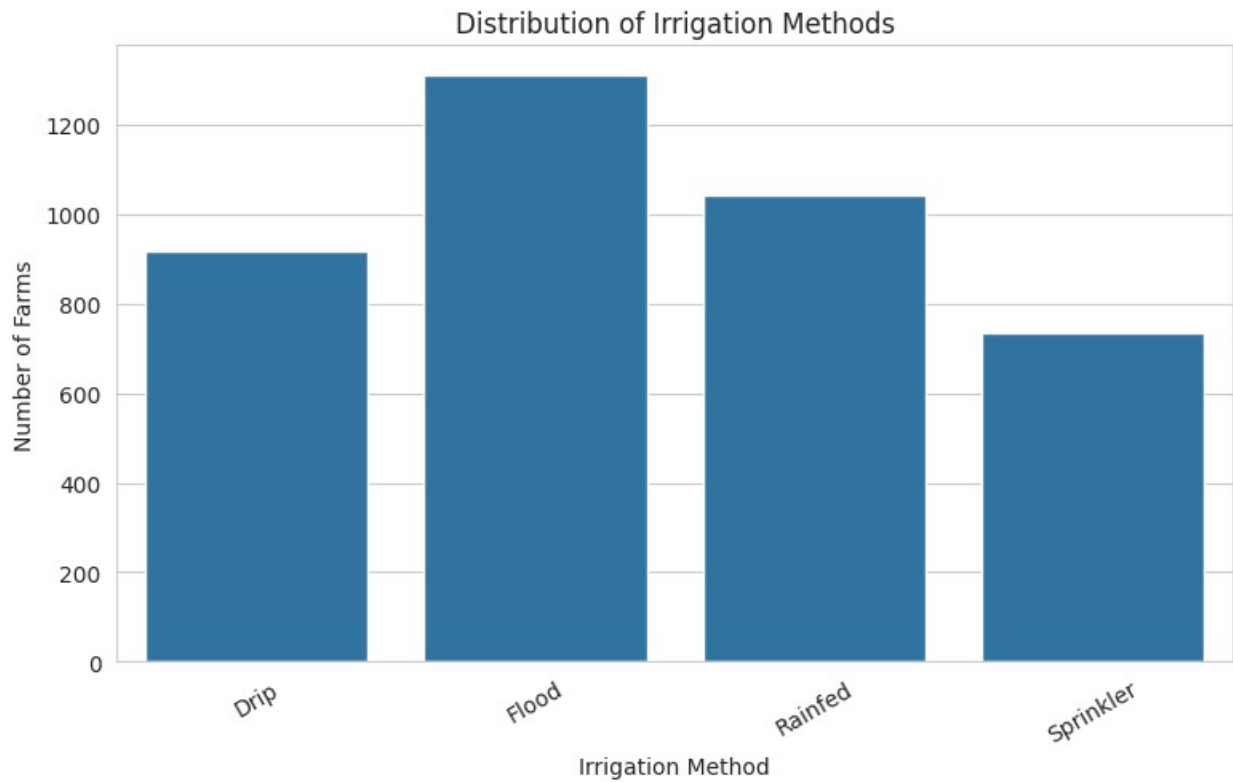
Kharif 28.454019

Rabi 23.489183

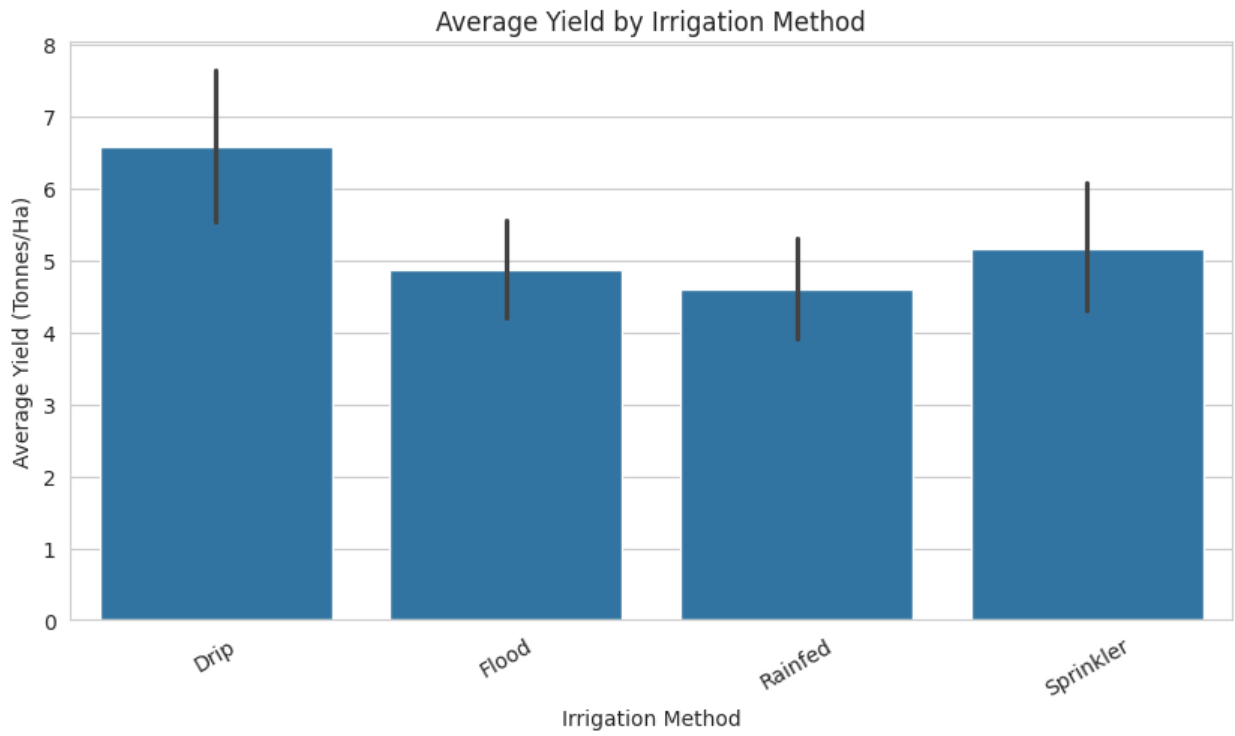
Name: Avg_Temperature_C, dtype: float64

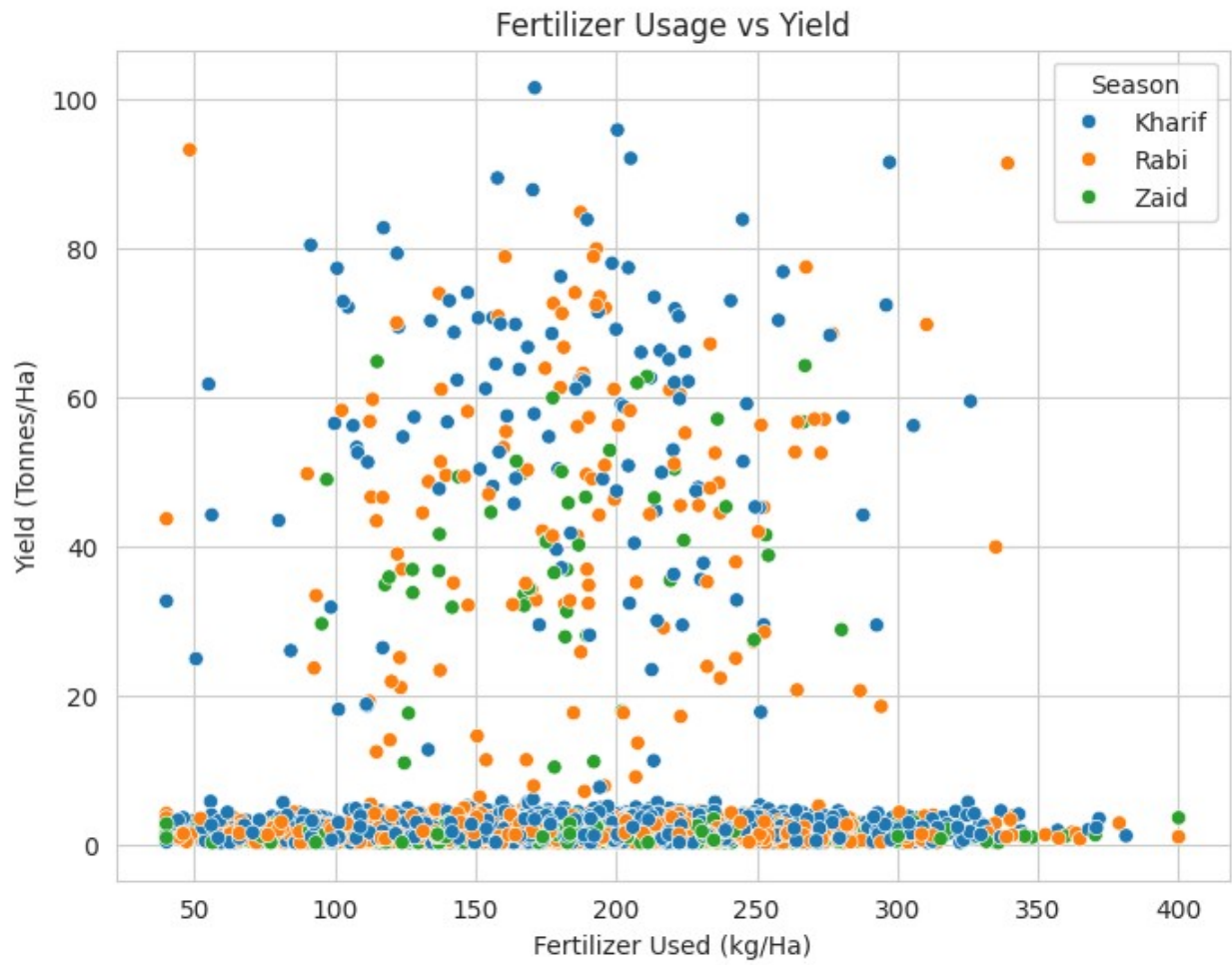


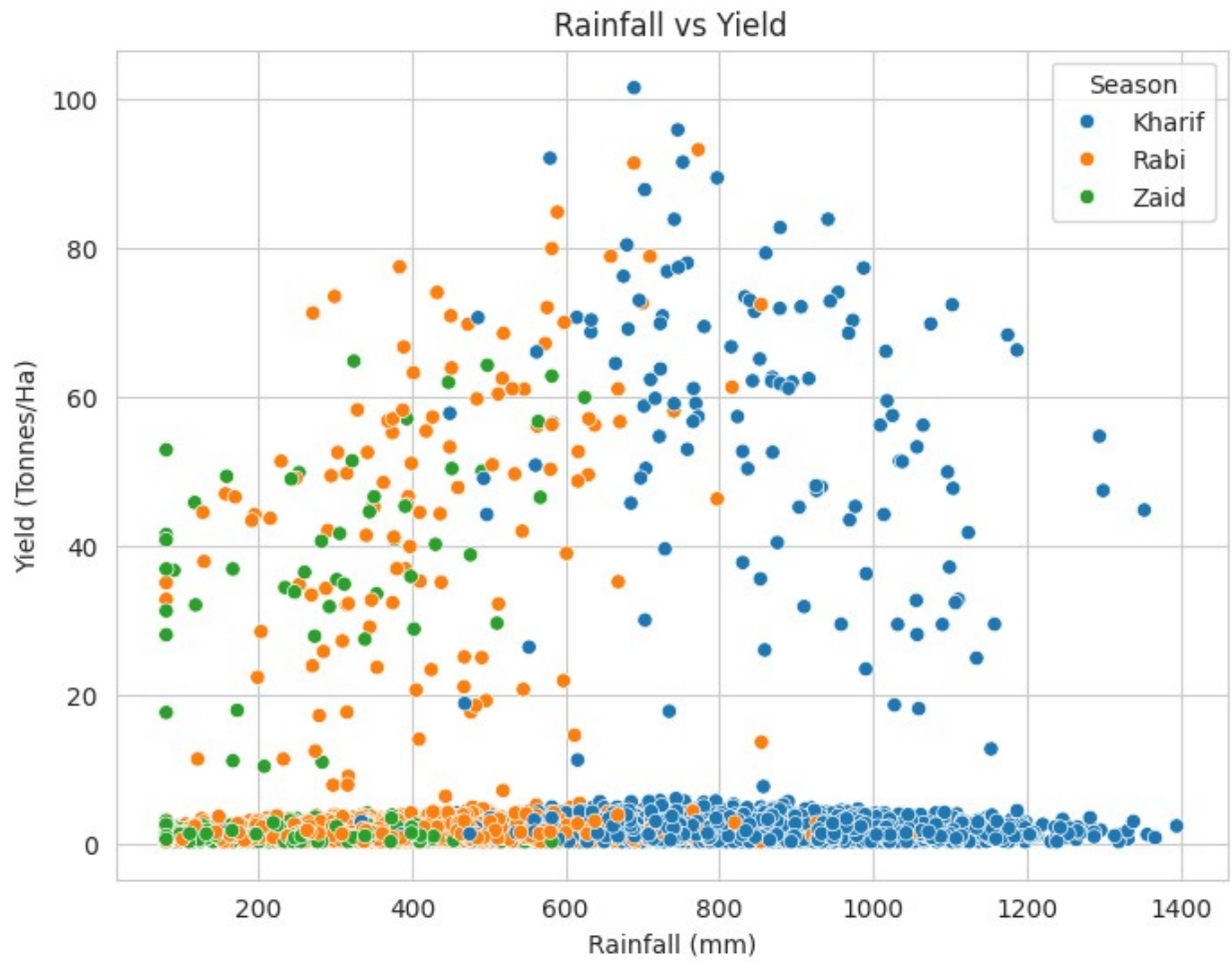
```
===== IRRIGATION METHOD COUNT =====  
Irrigation_Method  
Flood      1310  
Rainfed    1041  
Drip       915  
Sprinkler  734  
Name: count, dtype: int64
```

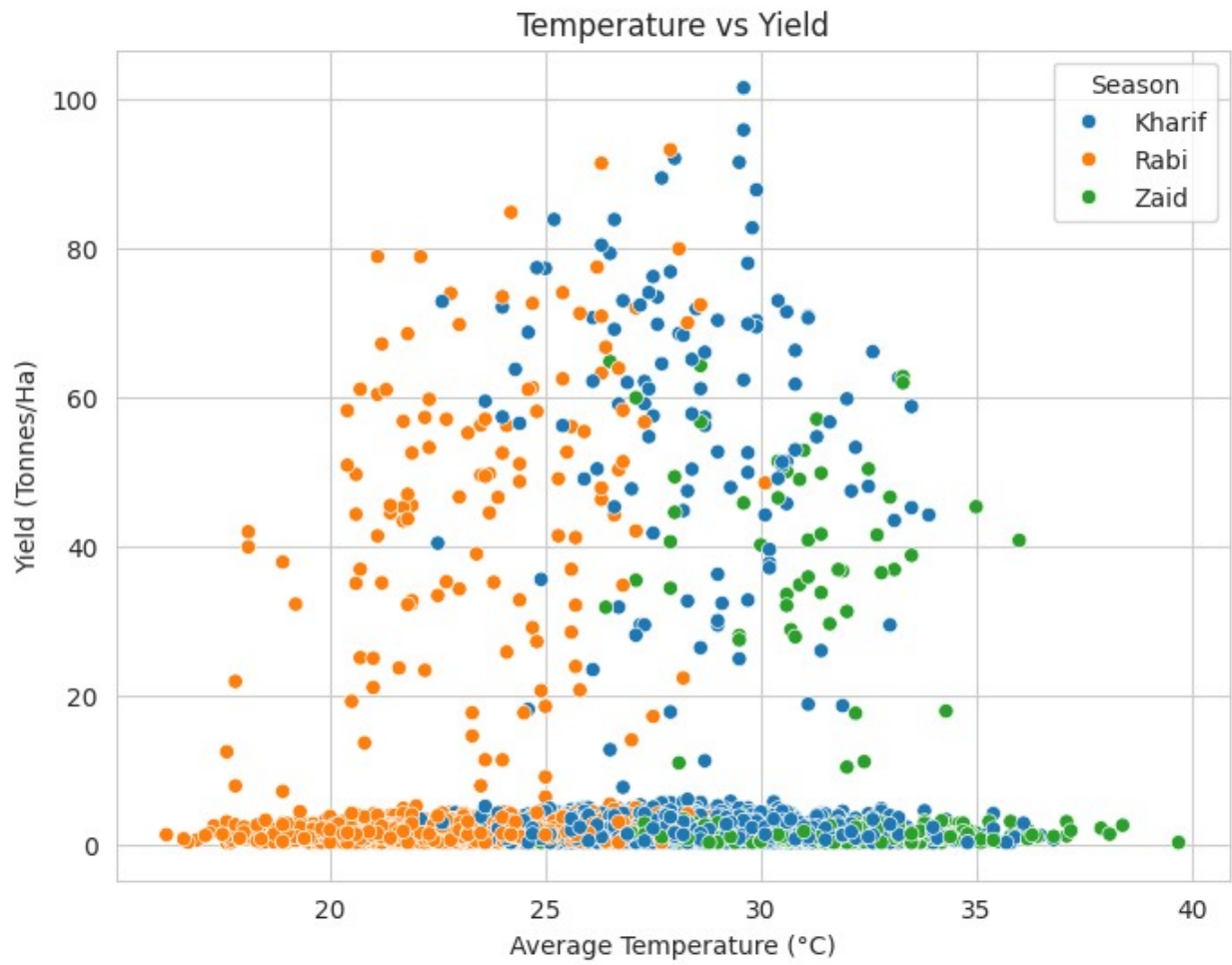


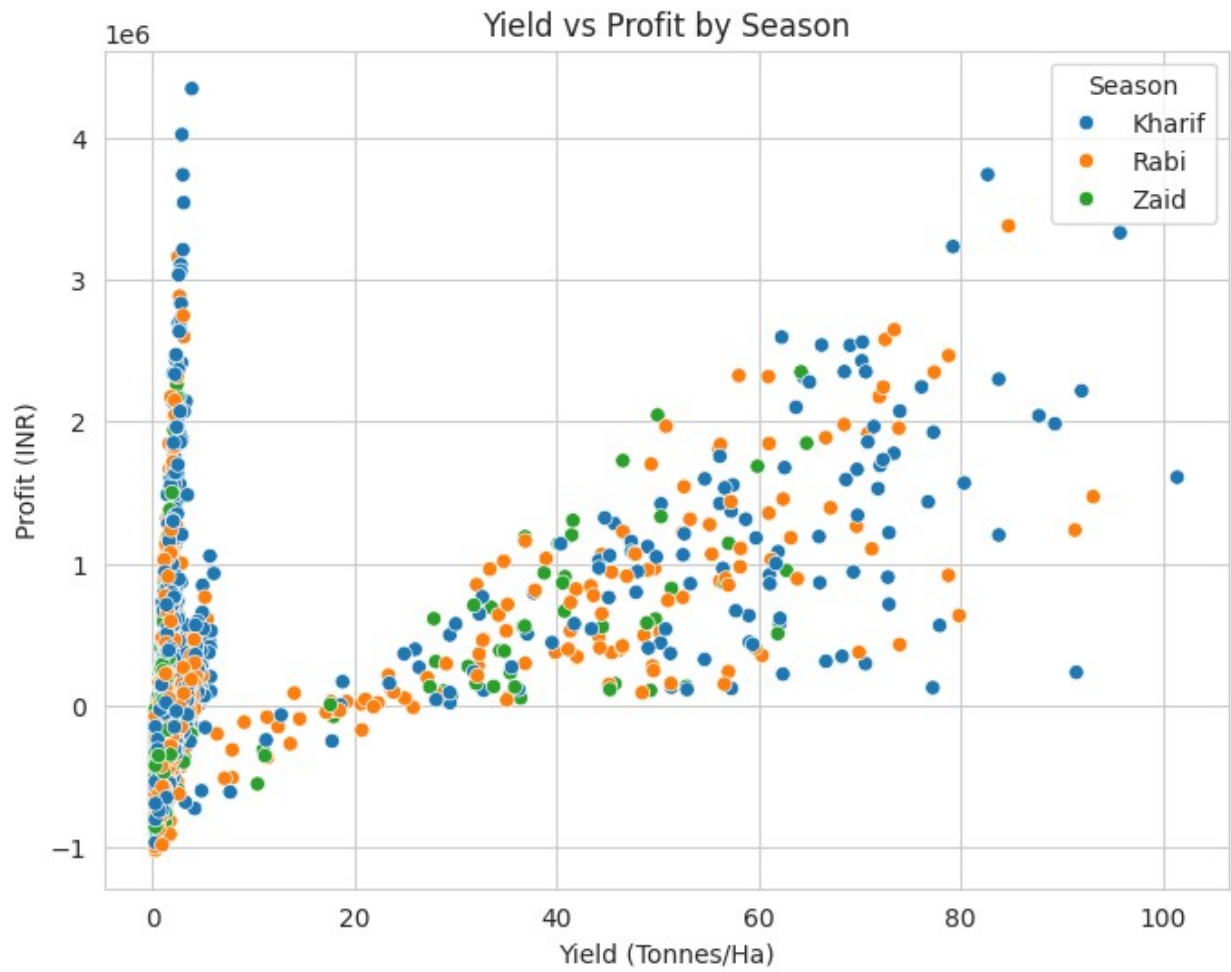
```
===== AVERAGE YIELD BY IRRIGATION METHOD =====  
Irrigation_Method  
Drip          6.577989  
Sprinkler     5.160627  
Flood         4.863977  
Rainfed       4.602142  
Name: Yield_Tonnes_Ha, dtype: float64
```

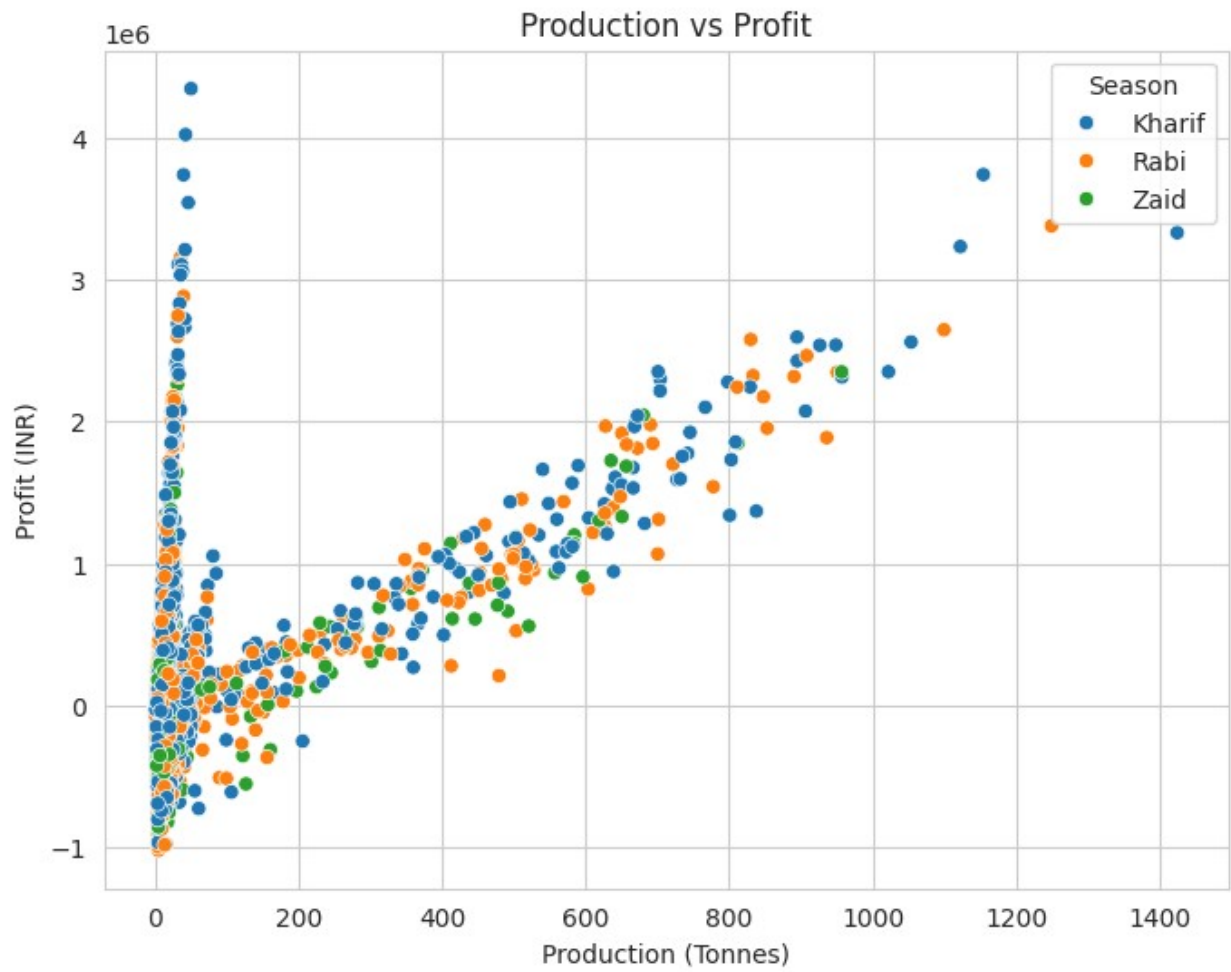














===== CORRELATION MATRIX =====

	Farm_Area_Hectares	Rainfall_mm \
Farm_Area_Hectares	1.000000	0.007256
Rainfall_mm	0.007256	1.000000
Avg_Temperature_C	0.036870	0.198492
Humidity_pct	0.000471	0.521069
Sunlight_Hours_Day	0.001821	-0.362779
Soil_pH	0.009857	0.016657
Soil_Moisture_pct	0.004033	0.508941
Nitrogen_kg_ha	0.013658	0.007460
Phosphorus_kg_ha	-0.006625	0.020940
Potassium_kg_ha	0.000195	0.031373
Fertilizer_kg_ha	0.014975	0.024602
Pesticide_Litre_ha	-0.005207	0.004058
Seed_Quality_Score	0.015236	-0.018581
Yield_Tonnes_Ha	0.030203	0.030951
Production_Tonnes	0.197842	0.024447
Market_Price_INR_Tonne	-0.021571	0.000639
Total_Cost_INR	0.962211	0.013041

Revenue_INR	0.543364	0.098146
Profit_INR	0.115241	0.107648
Water_Used_m3	0.576015	0.002240
Water_Efficiency_t_per_1000m3	0.028930	0.058102
Disease_Pest_Risk_pct	0.025963	0.621762
	Avg_Temperature_C	Humidity_pct \
Farm_Area_Hectares	0.036870	0.000471
Rainfall_mm	0.198492	0.521069
Avg_Temperature_C	1.000000	0.138136
Humidity_pct	0.138136	1.000000
Sunlight_Hours_Day	-0.033863	-0.297298
Soil_pH	0.029988	0.028447
Soil_Moisture_pct	0.084001	0.446470
Nitrogen_kg_ha	0.018707	0.001198
Phosphorus_kg_ha	0.008508	-0.005963
Potassium_kg_ha	0.022993	-0.003070
Fertilizer_kg_ha	0.015861	0.024623
Pesticide_Litre_ha	0.006657	-0.004028
Seed_Quality_Score	-0.031943	-0.045598
Yield_Tonnes_Ha	0.009996	0.012297
Production_Tonnes	0.007346	0.003614
Market_Price_INR_Tonne	0.002951	0.000262
Total_Cost_INR	0.041030	0.006023
Revenue_INR	0.026703	0.057358
Profit_INR	0.009043	0.063775
Water_Used_m3	0.022995	-0.002086
Water_Efficiency_t_per_1000m3	0.006534	0.026169
Disease_Pest_Risk_pct	0.245613	0.545093
	Sunlight_Hours_Day	Soil_pH \
Farm_Area_Hectares	0.001821	0.009857
Rainfall_mm	-0.362779	0.016657
Avg_Temperature_C	-0.033863	0.029988
Humidity_pct	-0.297298	0.028447
Sunlight_Hours_Day	1.000000	-0.017806
Soil_pH	-0.017806	1.000000
Soil_Moisture_pct	-0.302846	0.031183
Nitrogen_kg_ha	0.009920	0.006286
Phosphorus_kg_ha	0.001544	0.016620
Potassium_kg_ha	-0.014407	-0.019407
Fertilizer_kg_ha	-0.013073	-0.010591
Pesticide_Litre_ha	-0.017238	-0.010135
Seed_Quality_Score	0.021257	-0.004181
Yield_Tonnes_Ha	-0.015341	-0.020672
Production_Tonnes	-0.023504	-0.012512
Market_Price_INR_Tonne	-0.000696	0.022226
Total_Cost_INR	-0.000655	0.005302
Revenue_INR	-0.047216	-0.008798

Profit_INR	-0.054822	-0.013145
Water_Used_m3	-0.005717	-0.022859
Water_Efficiency_t_per_1000m3	-0.021386	-0.010087
Disease_Pest_Risk_pct	-0.253251	0.031141
	Soil_Moisture_pct	Nitrogen_kg_ha \
Farm_Area_Hectares	0.004033	0.013658
Rainfall_mm	0.508941	0.007460
Avg_Temperature_C	0.084001	0.018707
Humidity_pct	0.446470	0.001198
Sunlight_Hours_Day	-0.302846	0.009920
Soil_pH	0.031183	0.006286
Soil_Moisture_pct	1.000000	0.008884
Nitrogen_kg_ha	0.008884	1.000000
Phosphorus_kg_ha	0.013406	0.010199
Potassium_kg_ha	0.007796	0.010490
Fertilizer_kg_ha	0.008782	-0.032440
Pesticide_Litre_ha	-0.023159	0.014280
Seed_Quality_Score	-0.019556	0.014896
Yield_Tonnes_Ha	0.010999	0.053149
Production_Tonnes	0.004787	0.048182
Market_Price_INR_Tonne	0.016914	-0.022595
Total_Cost_INR	0.002741	0.011900
Revenue_INR	0.079380	0.078252
Profit_INR	0.091281	0.085016
Water_Used_m3	0.003183	0.022127
Water_Efficiency_t_per_1000m3	0.027776	0.054896
Disease_Pest_Risk_pct	0.375547	-0.011503
	Phosphorus_kg_ha	Potassium_kg_ha \
Farm_Area_Hectares	-0.006625	0.000195
Rainfall_mm	0.020940	0.031373
Avg_Temperature_C	0.008508	0.022993
Humidity_pct	-0.005963	-0.003070
Sunlight_Hours_Day	0.001544	-0.014407
Soil_pH	0.016620	-0.019407
Soil_Moisture_pct	0.013406	0.007796
Nitrogen_kg_ha	0.010199	0.010490
Phosphorus_kg_ha	1.000000	0.002847
Potassium_kg_ha	0.002847	1.000000
Fertilizer_kg_ha	0.012905	0.017254
Pesticide_Litre_ha	-0.029942	-0.004675
Seed_Quality_Score	-0.007575	0.006862
Yield_Tonnes_Ha	0.047941	0.003382
Production_Tonnes	0.034957	0.004472
Market_Price_INR_Tonne	-0.005174	0.007572
Total_Cost_INR	-0.016508	0.003431
Revenue_INR	0.034940	0.028278
Profit_INR	0.049747	0.031192

Water_Used_m3	-0.000752	0.001667
Water_Efficiency_t_per_1000m3	0.057129	0.006575
Disease_Pest_Risk_pct	0.032061	0.013403
	Fertilizer_kg_ha	Pesticide_Litre_ha \
Farm_Area_Hectares	0.014975	-0.005207
Rainfall_mm	0.024602	0.004058
Avg_Temperature_C	0.015861	0.006657
Humidity_pct	0.024623	-0.004028
Sunlight_Hours_Day	-0.013073	-0.017238
Soil_pH	-0.010591	-0.010135
Soil_Moisture_pct	0.008782	-0.023159
Nitrogen_kg_ha	-0.032440	0.014280
Phosphorus_kg_ha	0.012905	-0.029942
Potassium_kg_ha	0.017254	-0.004675
Fertilizer_kg_ha	1.000000	-0.001178
Pesticide_Litre_ha	-0.001178	1.000000
Seed_Quality_Score	0.016144	-0.001462
Yield_Tonnes_Ha	0.000018	-0.007779
Production_Tonnes	0.001023	-0.009021
Market_Price_INR_Tonne	-0.000962	0.007342
Total_Cost_INR	0.147353	0.046560
Revenue_INR	0.005176	-0.000476
Profit_INR	-0.073542	-0.025705
Water_Used_m3	0.000895	0.002177
Water_Efficiency_t_per_1000m3	0.005077	-0.006330
Disease_Pest_Risk_pct	0.017851	-0.465183
	Seed_Quality_Score	Yield_Tonnes_Ha \
Farm_Area_Hectares	0.015236	0.030203
Rainfall_mm	-0.018581	0.030951
Avg_Temperature_C	-0.031943	0.009996
Humidity_pct	-0.045598	0.012297
Sunlight_Hours_Day	0.021257	-0.015341
Soil_pH	-0.004181	-0.020672
Soil_Moisture_pct	-0.019556	0.010999
Nitrogen_kg_ha	0.014896	0.053149
Phosphorus_kg_ha	-0.007575	0.047941
Potassium_kg_ha	0.006862	0.003382
Fertilizer_kg_ha	0.016144	0.000018
Pesticide_Litre_ha	-0.001462	-0.007779
Seed_Quality_Score	1.000000	-0.007209
Yield_Tonnes_Ha	-0.007209	1.000000
Production_Tonnes	-0.000387	0.882813
Market_Price_INR_Tonne	0.011960	-0.382607
Total_Cost_INR	0.022033	0.030622
Revenue_INR	0.019116	0.432152
Profit_INR	0.010438	0.488466
Water_Used_m3	0.017020	0.386294

Water_Efficiency_t_per_1000m3	-0.012031	0.912568
Disease_Pest_Risk_pct	-0.024602	0.014292

Production_Tonnes

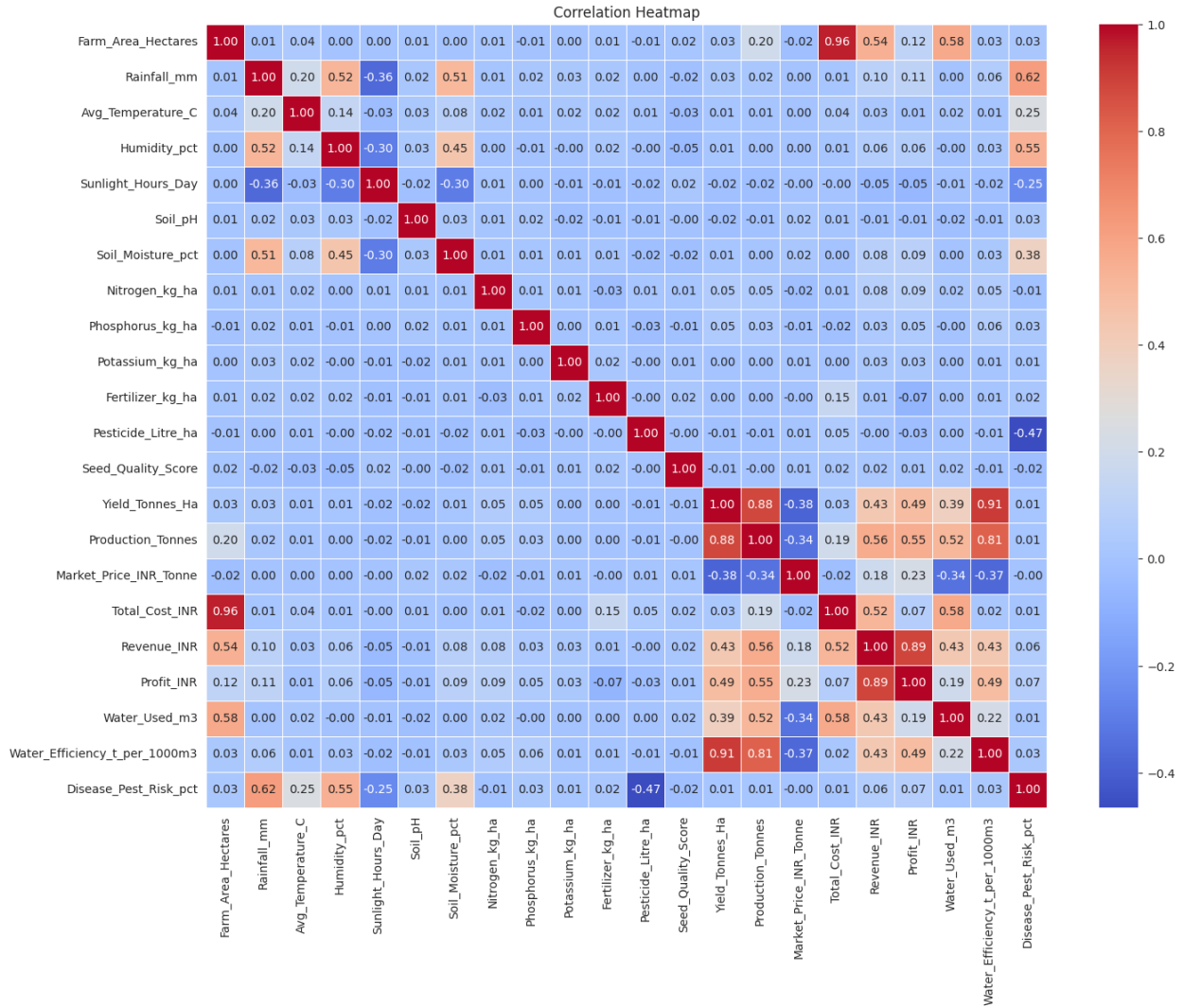
Market_Price_INR_Tonne \		
Farm_Area_Hectares	0.197842	-
0.021571		
Rainfall_mm	0.024447	
0.000639		
Avg_Temperature_C	0.007346	
0.002951		
Humidity_pct	0.003614	
0.000262		
Sunlight_Hours_Day	-0.023504	-
0.000696		
Soil_pH	-0.012512	
0.022226		
Soil_Moisture_pct	0.004787	
0.016914		
Nitrogen_kg_ha	0.048182	-
0.022595		
Phosphorus_kg_ha	0.034957	-
0.005174		
Potassium_kg_ha	0.004472	
0.007572		
Fertilizer_kg_ha	0.001023	-
0.000962		
Pesticide_Litre_ha	-0.009021	
0.007342		
Seed_Quality_Score	-0.000387	
0.011960		
Yield_Tonnes_Ha	0.882813	-
0.382607		
Production_Tonnes	1.000000	-
0.337531		
Market_Price_INR_Tonne	-0.337531	
1.000000		
Total_Cost_INR	0.193158	-
0.022294		
Revenue_INR	0.563505	
0.182338		
Profit_INR	0.554172	
0.225119		
Water_Used_m3	0.515557	-
0.336883		
Water_Efficiency_t_per_1000m3	0.812143	-
0.372592		
Disease_Pest_Risk_pct	0.013961	-
0.000454		

	Total_Cost_INR	Revenue_INR	Profit_INR
\			
Farm_Area_Hectares	0.962211	0.543364	0.115241
Rainfall_mm	0.013041	0.098146	0.107648
Avg_Temperature_C	0.041030	0.026703	0.009043
Humidity_pct	0.006023	0.057358	0.063775
Sunlight_Hours_Day	-0.000655	-0.047216	-0.054822
Soil_pH	0.005302	-0.008798	-0.013145
Soil_Moisture_pct	0.002741	0.079380	0.091281
Nitrogen_kg_ha	0.011900	0.078252	0.085016
Phosphorus_kg_ha	-0.016508	0.034940	0.049747
Potassium_kg_ha	0.003431	0.028278	0.031192
Fertilizer_kg_ha	0.147353	0.005176	-0.073542
Pesticide_Litre_ha	0.046560	-0.000476	-0.025705
Seed_Quality_Score	0.022033	0.019116	0.010438
Yield_Tonnes_Ha	0.030622	0.432152	0.488466
Production_Tonnes	0.193158	0.563505	0.554172
Market_Price_INR_Tonne	-0.022294	0.182338	0.225119
Total_Cost_INR	1.000000	0.520709	0.068356
Revenue_INR	0.520709	1.000000	0.887331
Profit_INR	0.068356	0.887331	1.000000
Water_Used_m3	0.575978	0.429053	0.190278
Water_Efficiency_t_per_1000m3	0.020402	0.428302	0.489487
Disease_Pest_Risk_pct	0.005922	0.060571	0.067583
Water_Efficiency_t_per_1000m3	Water_Used_m3		
Farm_Area_Hectares	\		
	0.576015		

0.028930		
Rainfall_mm	0.002240	
0.058102		
Avg_Temperature_C	0.022995	
0.006534		
Humidity_pct	-0.002086	
0.026169		
Sunlight_Hours_Day	-0.005717	-
0.021386		
Soil_pH	-0.022859	-
0.010087		
Soil_Moisture_pct	0.003183	
0.027776		
Nitrogen_kg_ha	0.022127	
0.054896		
Phosphorus_kg_ha	-0.000752	
0.057129		
Potassium_kg_ha	0.001667	
0.006575		
Fertilizer_kg_ha	0.000895	
0.005077		
Pesticide_Litre_ha	0.002177	-
0.006330		
Seed_Quality_Score	0.017020	-
0.012031		
Yield_Tonnes_Ha	0.386294	
0.912568		
Production_Tonnes	0.515557	
0.812143		
Market_Price_INR_Tonne	-0.336883	-
0.372592		
Total_Cost_INR	0.575978	
0.020402		
Revenue_INR	0.429053	
0.428302		
Profit_INR	0.190278	
0.489487		
Water_Used_m3	1.000000	
0.217081		
Water_Efficiency_t_per_1000m3	0.217081	
1.000000		
Disease_Pest_Risk_pct	0.011485	
0.029253		

	Disease_Pest_Risk_pct
Farm_Area_Hectares	0.025963
Rainfall_mm	0.621762
Avg_Temperature_C	0.245613
Humidity_pct	0.545093

Sunlight_Hours_Day	-0.253251
Soil_pH	0.031141
Soil_Moisture_pct	0.375547
Nitrogen_kg_ha	-0.011503
Phosphorus_kg_ha	0.032061
Potassium_kg_ha	0.013403
Fertilizer_kg_ha	0.017851
Pesticide_Litre_ha	-0.465183
Seed_Quality_Score	-0.024602
Yield_Tonnes_Ha	0.014292
Production_Tonnes	0.013961
Market_Price_INR_Tonne	-0.000454
Total_Cost_INR	0.005922
Revenue_INR	0.060571
Profit_INR	0.067583
Water_Used_m3	0.011485
Water_Efficiency_t_per_1000m3	0.029253
Disease_Pest_Risk_pct	1.000000



===== CORRELATION WITH PROFIT =====

```

Profit_INR                1.000000
Revenue_INR               0.887331
Production_Tonnes         0.554172
Water_Efficiency_t_per_1000m3 0.489487
Yield_Tonnes_Ha           0.488466
Market_Price_INR_Tonne    0.225119
Water_Used_m3             0.190278
Farm_Area_Hectares        0.115241
Rainfall_mm               0.107648
Soil_Moisture_pct         0.091281
Nitrogen_kg_ha            0.085016
Total_Cost_INR            0.068356
Disease_Pest_Risk_pct     0.067583
Humidity_pct              0.063775
Phosphorus_kg_ha          0.049747
  
```

```

Potassium_kg_ha      0.031192
Seed_Quality_Score   0.010438
Avg_Temperature_C    0.009043
Soil_pH              -0.013145
Pesticide_Litre_ha   -0.025705
Sunlight_Hours_Day   -0.054822
Fertilizer_kg_ha     -0.073542
Name: Profit_INR, dtype: float64

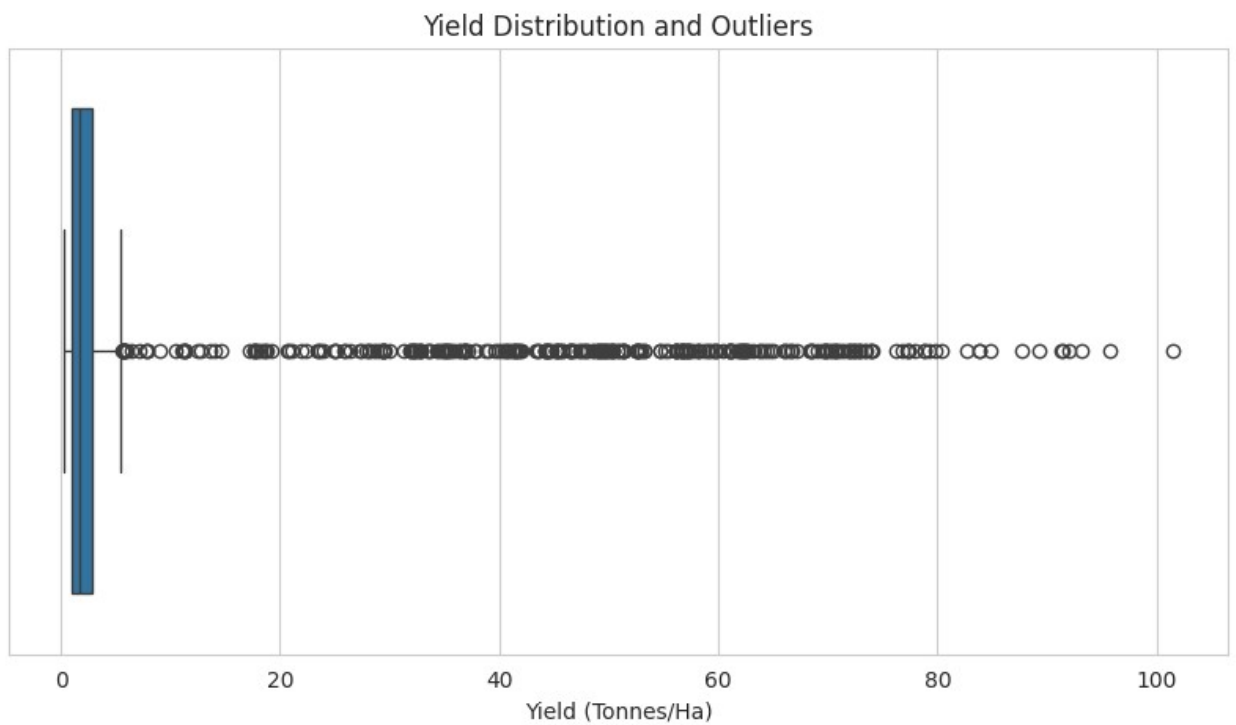
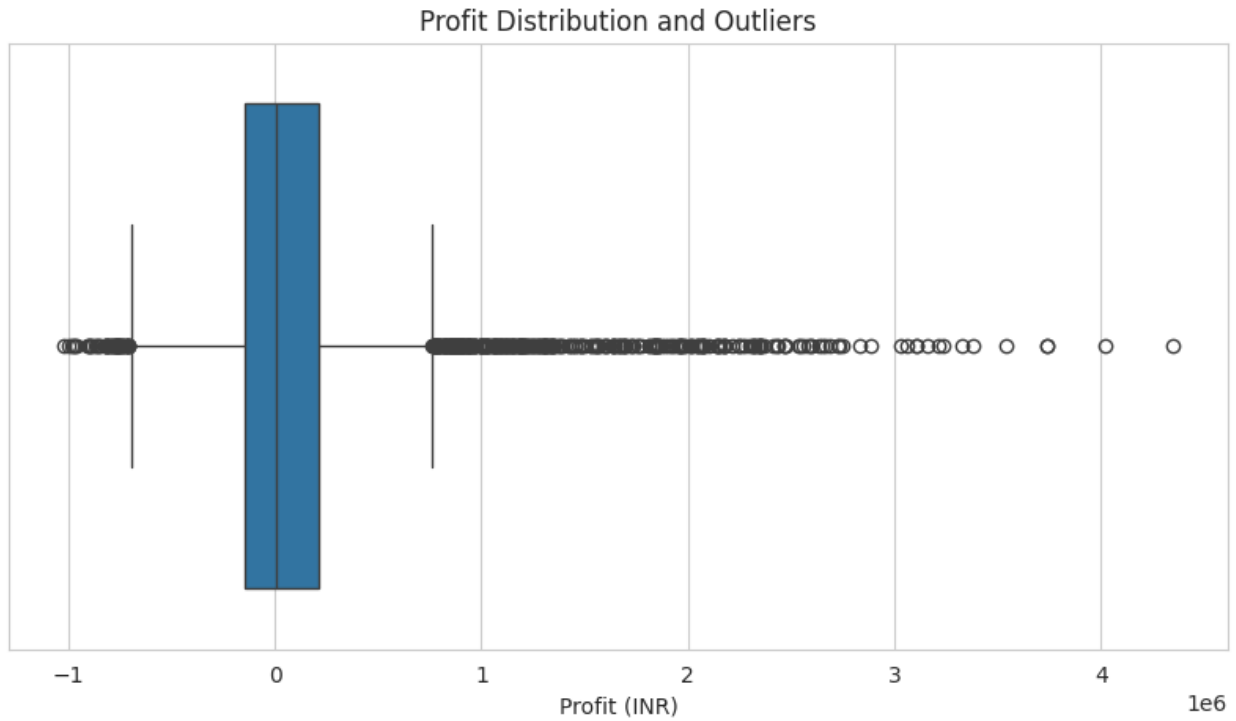
```

===== TOP 10 FARMS BY PROFIT =====

	Farm_ID	State	Crop	Season	Profit_INR
2236	SF12237	Punjab	Chilli	Kharif	4345421
626	SF10627	Maharashtra	Chilli	Kharif	4021850
3425	SF13426	Madhya Pradesh	Sugarcane	Kharif	3740099
1690	SF11691	Karnataka	Chilli	Kharif	3738329
247	SF10248	Maharashtra	Chilli	Kharif	3542204
357	SF10358	Punjab	Sugarcane	Rabi	3378879
3423	SF13424	Andhra Pradesh	Sugarcane	Kharif	3330582
1930	SF11931	Tamil Nadu	Sugarcane	Kharif	3232970
3983	SF13984	Gujarat	Chilli	Kharif	3211345
263	SF10264	Maharashtra	Chilli	Rabi	3159447

===== BOTTOM 10 FARMS BY PROFIT =====

	Farm_ID	State	Crop	Season	Profit_INR
780	SF10781	Andhra Pradesh	Groundnut	Rabi	-1019223
2210	SF12211	Andhra Pradesh	Wheat	Rabi	-995195
3625	SF13626	Punjab	Maize	Rabi	-981050
2182	SF12183	Andhra Pradesh	Rice	Kharif	-976860
104	SF10105	Gujarat	Wheat	Zaid	-965197
3394	SF13395	Gujarat	Wheat	Kharif	-963396
2730	SF12731	Madhya Pradesh	Rice	Rabi	-906082
1311	SF11312	Punjab	Rice	Zaid	-902740
585	SF10586	Telangana	Cotton	Zaid	-895730
3090	SF13091	Karnataka	Maize	Kharif	-890641



```

===== COMPLETE SEASONAL SUMMARY =====
      Farms Avg_Yield Avg_Production Avg_Revenue Avg_Cost
Avg_Profit \
Season
  
```

Kharif	1779	5.63	46.31	710719.06	531804.41	
178914.65						
Rabi	1627	5.04	41.49	601526.05	513836.58	
87689.47						
Zaid	594	4.64	38.89	519171.90	543976.73	-
24804.82						

	Avg_Water_Used	Avg_Water_Efficiency	Avg_Disease_Risk
Season			
Kharif	6102.20	5.89	54.47
Rabi	5846.99	5.19	40.48
Zaid	6419.89	4.41	38.22

===== BEST PERFORMING SEASON =====

Best Season: Kharif

Average Profit: 178914.65

===== LOWEST PERFORMING SEASON =====

Lowest Season: Zaid

Average Profit: -24804.82

===== MOST PROFITABLE CROP =====

Crop: Sugarcane

Average Profit: 817187.9868852459

===== HIGHEST YIELD CROP =====

Crop: Sugarcane

Average Yield: 46.642360655737704

===== HIGHEST PROFIT STATE =====

State: Punjab

Average Profit: 136025.63636363635

===== MOST WATER-EFFICIENT CROP =====

Crop: Sugarcane

Water Efficiency: 30.477672131147543

===== HIGHEST DISEASE/PEST RISK SEASON =====

Season: Kharif

Average Risk: 54.46790331646992

===== HIGHEST WATER USAGE SEASON =====

Season: Zaid

Average Water Usage: 6419.893939393939

=====

FINAL KEY FINDINGS

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1. Best performing season based on profit: Kharif
Average profit: ₹178,914.65
2. Lowest performing season based on profit: Zaid
Average profit: ₹-24,804.82
3. Most profitable crop: Sugarcane
Average profit: ₹817,187.99
4. Highest yield crop: Sugarcane
Average yield: 46.64 tonnes/ha
5. Highest profit state: Punjab
Average profit: ₹136,025.64
6. Most water-efficient crop: Sugarcane
Water efficiency: 30.48
7. Highest disease/pest risk season: Kharif
Average risk: 54.47%
8. Highest water usage season: Zaid
Average water usage: 6,419.89 m³

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RECOMMENDATIONS

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Analysis completed successfully!
Summary files have been saved as CSV files.